

Digestion

Ryan Christopher G. Atuel, RPh, MSc

Department of Pharmacy
College of Nursing and Allied Health Sciences
Cebu Institute of Technology University
Cebu, Philippines



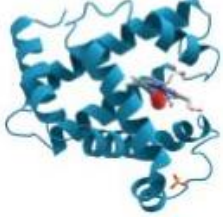
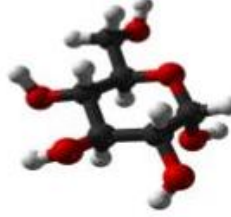
Digestion and Absorption

- Digestion is the **chemical breakdown of large food molecules into smaller** molecules that can be used by cells.
- The breakdown occurs **when certain specific enzymes are mixed** with the food.



Major Dietary Components

- **Carbohydrates** → absorbed as monosaccharides
- **Lipids/Fats** → absorbed as free fatty acids + glycerol
- **Proteins** → absorbed as amino acids + small peptides

Proteins	Carbohydrates	Lipids
		
		

“Digestion = hydrolysis → smaller absorbable units”

Carbohydrates Digestion

- Carbohydrates from food (starch, glycogen, sugars) must be **broken down into monosaccharides** for absorption.

Main sites of digestion:

1. **Mouth** – partial digestion by *salivary amylase*
2. **Small intestine** – major digestion by *pancreatic amylase*
3. **Intestinal mucosa** – final digestion to monosaccharides

💡 **End products:** Glucose, Galactose, Fructose → absorbed into blood.



Carbohydrate Digestion

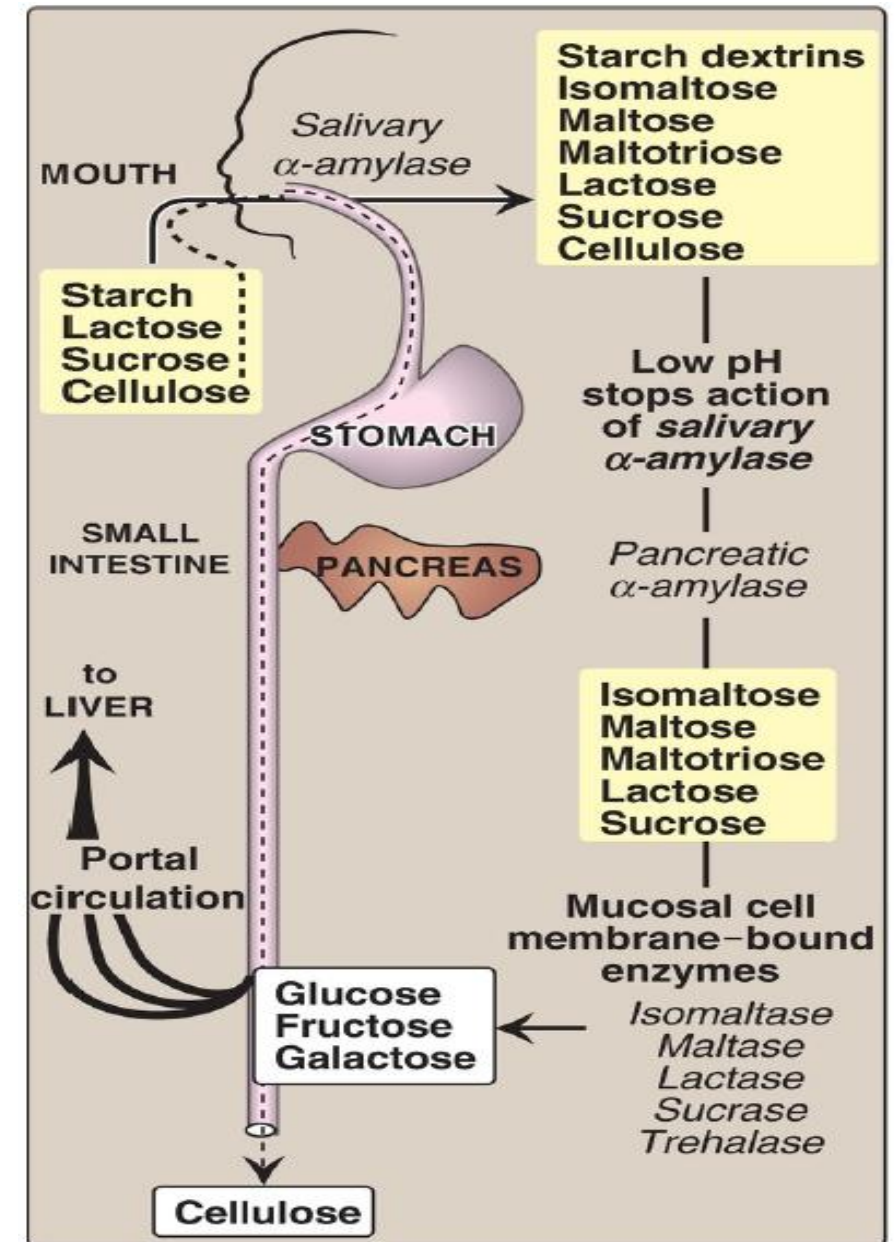
Enzyme: *Salivary α -amylase*

- Hydrolyzes **α -glycosidic bonds** in starch (plants) & glycogen (meats).

- Produces: **maltose + smaller polysaccharides**

- Only *brief digestion* in mouth $\rightarrow$ food swallowed quickly.

- In the stomach:** no carbohydrate digestion (amylase inactivated by acid).



Carbohydrate Digestion

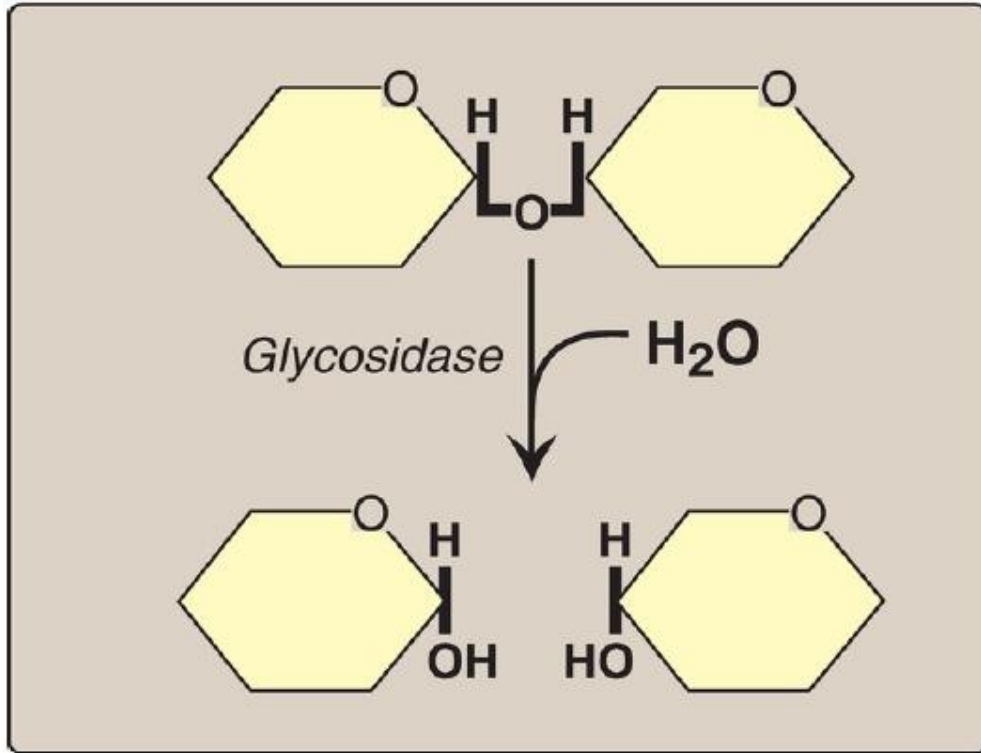


Figure 7.8 Hydrolysis of a glycosidic bond.

Principal Sites:

- **Mouth:** salivary α -amylase begins starch & glycogen hydrolysis.

- **Intestinal lumen:** pancreatic α -amylase continues digestion to maltose & small oligosaccharides.

Enzymes: Glycoside Hydrolases (Glycosidases)

- Catalyze hydrolysis of **glycosidic bonds** in carbohydrates.

Types:

- **Endoglycosidases** – cleave internal bonds in polysaccharides (e.g., α -amylase).
- **Disaccharidases** – act on di- and trisaccharides (e.g., maltase, sucrase, lactase).

Glycosidases are **specific** for sugar structure, bond configuration (α or β), and linkage type.
— > explains why **amylase cannot digest cellulose (β -1,4 bonds)**.

Carbohydrate Digestion

Polysaccharides

1. Starch is a major plant storage form of glucose. It occurs in two forms:
 - **alpha-amylase** = glucoses are linked together in straight chains
 - **Amylopectin** = glucose chains are highly branched
 - Note: Except for the branch points of amylopectin,
 - the glucose monomers in starch are linked via **alpha(1->4) glycosidic bonds**, which, in the digestive tract of mammals, are **hydrolyzed by amylases**.
2. Dietary fibers (Cellulose) is the other major plant carbohydrate.
 - Major constituent of plant cell walls
 - Cellulose is composed on unbranched, linear chains of D-glucose molecules, linked to one another by beta(1->4) glycosidic bonds
 - no vertebrate has the capacity to enzymatically digest

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Carbohydrate Digestion

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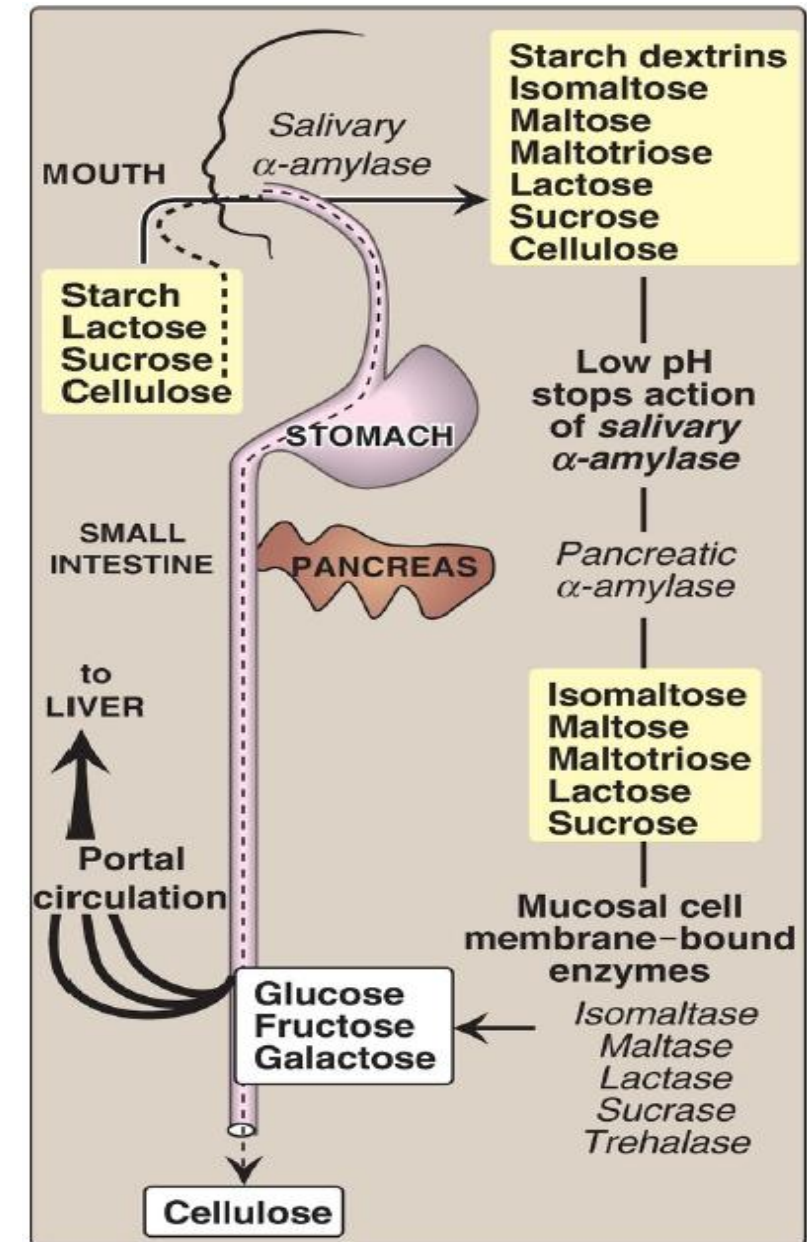
Carbohydrate Digestion

1. Salivary α -Amylase

- Acts briefly during **mastication**.
 - Hydrolyzes random $\alpha(1\rightarrow4)$ bonds in **starch (amylose, amylopectin)** and **glycogen**.
 - Cannot hydrolyze $\alpha(1\rightarrow6)$ or $\beta(1\rightarrow4)$ bonds $\rightarrow$ **dextrins** (short branched chains) remain.
 - **Inactivated by stomach acid** $\rightarrow$ digestion pauses in the stomach.
- Humans lack $\beta(1\rightarrow4)$ -endoglucosidase $\rightarrow$ cannot digest cellulose.*

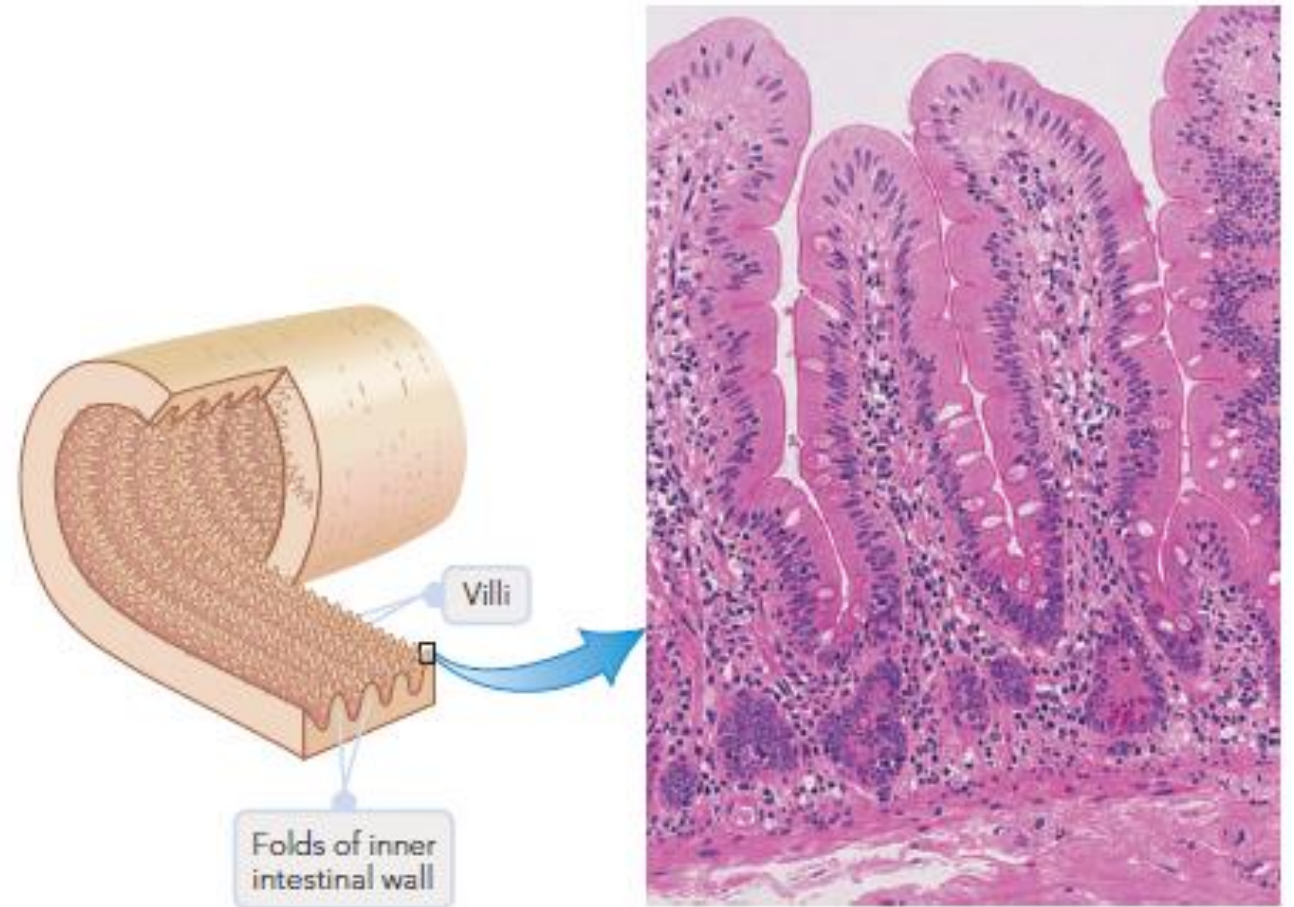
2. Pancreatic α -Amylase

- Active in **small intestine** after acid neutralization by **bicarbonate**.
- Continues digestion $\rightarrow$ produces **maltose, maltotriose, and isomaltose**.



Carbohydrate Digestion

The three major breakdown products from carbohydrate digestion are thus **glucose**, **galactose**, and **fructose**. These **monosaccharides** are absorbed **into the bloodstream through the intestinal wall**. The folds of the intestinal wall are lined with fingerlike projections called *villi*, which are rich in blood capillaries

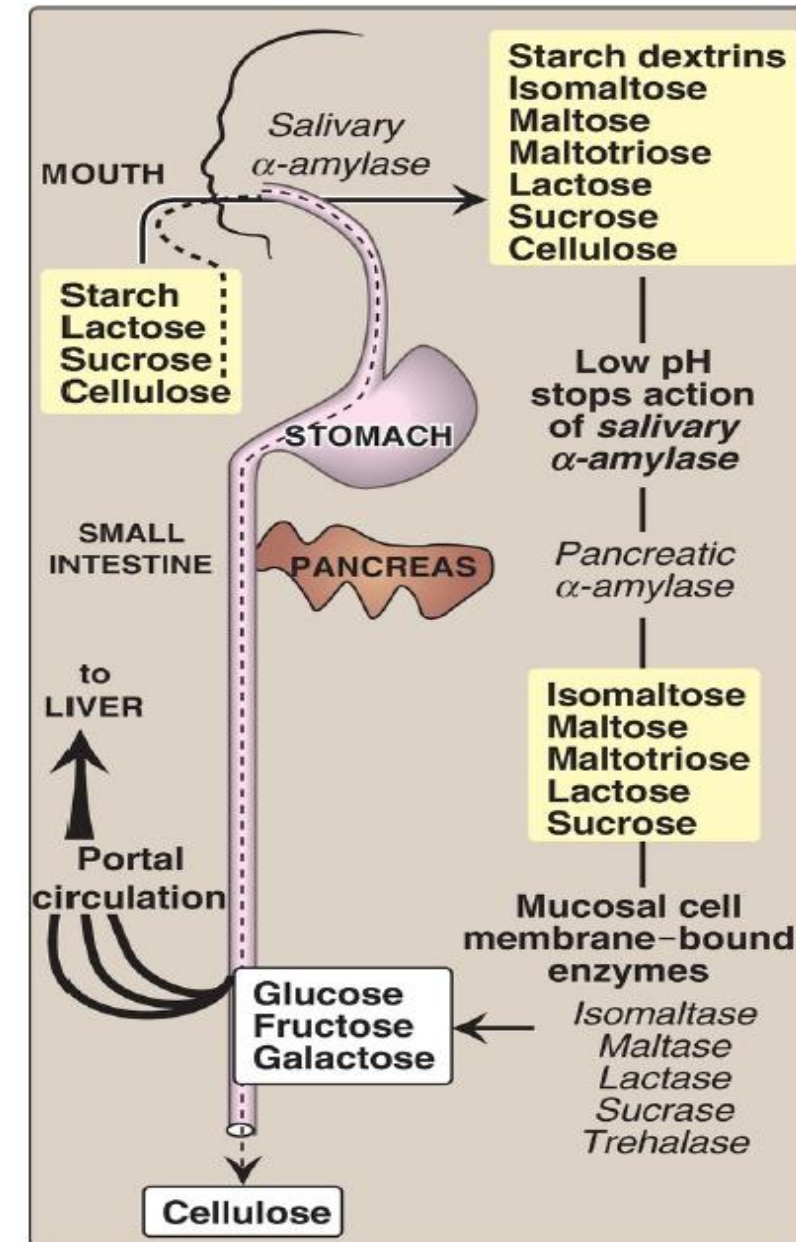


Carbohydrate Digestion

3. Intestinal Disaccharidases (Brush Border Enzymes)

Enzyme	Substrate	Bond	Products
Isomaltase	Isomaltose	$\alpha(1\rightarrow6)$	Glucose
Maltase	Maltose, Maltotriose	$\alpha(1\rightarrow4)$	Glucose
Sucrase	Sucrose	$\alpha(1\rightarrow2)$	Glucose + Fructose
Lactase	Lactose	$\beta(1\rightarrow4)$	Galactose + Glucose
Trehalase	Trehalose (from fungi)	$\alpha(1\rightarrow1)$	2 Glucose

- **Sucrase–Isomaltase (SI complex)** → single protein with two enzyme sites.
- **Maltase–Glucoamylase (MGA)** → one protein, two activities; glucoamylase acts on dextrins.



Carbohydrate Digestion

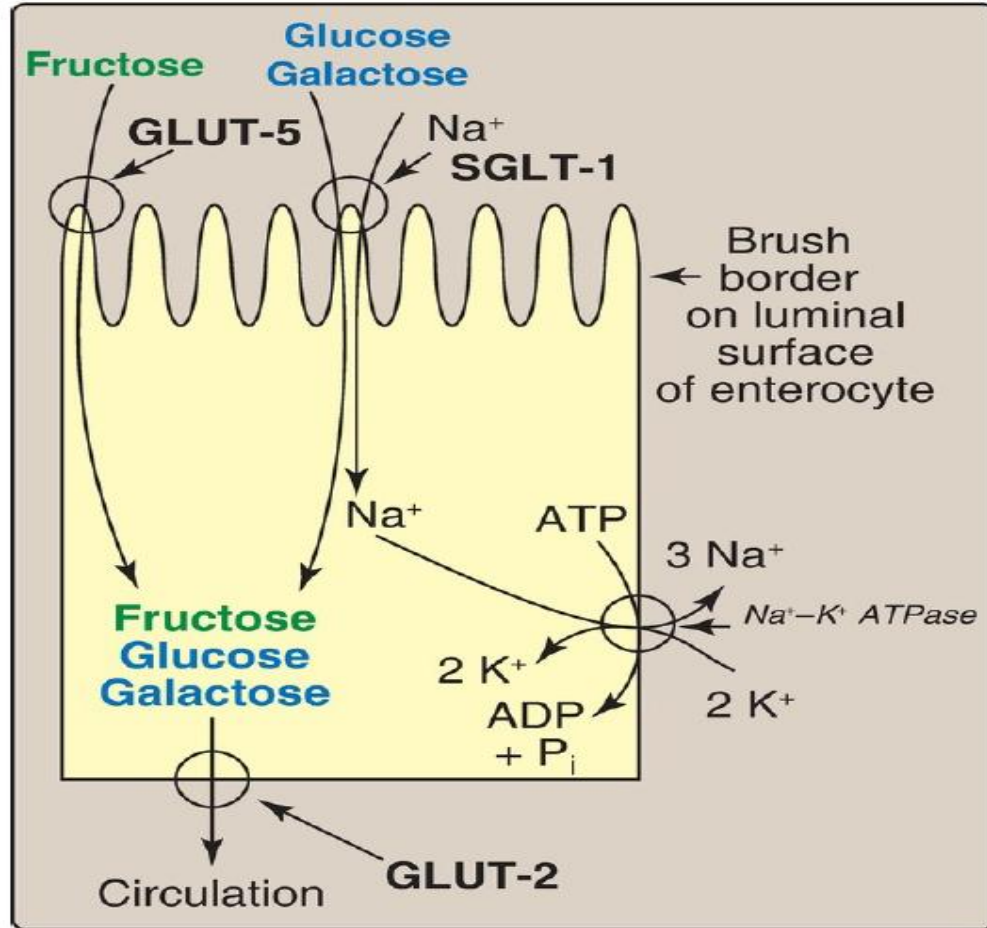


Figure 7.10 Absorption by enterocytes of the monosaccharide products of carbohydrate digestion. GLUT = glucose transporter; SGLT-1 = sodium (Na⁺)-dependent glucose cotransporter. K⁺ = potassium.

Absorption of Monosaccharide

Site: Upper jejunum (main site of sugar absorption).
Final Products: Glucose, Galactose, Fructose.

Mechanism Summary:

- **Na⁺/K⁺-ATPase** maintains gradient → drives SGLT-1 transport.
- Monosaccharides exit via **GLUT-2** → **portal circulation** → **liver**.
- In the **liver**, galactose & fructose → converted to glucose derivatives.

Absorption depends on **specific transporters**, not passive diffusion.

→ SGLT-1 (active), GLUT-5 (facilitated), GLUT-2 (exit).

Carbohydrate Digestion

Abnormal Degradation of Disaccharides

Normal digestion:

Efficient digestion → All carbohydrates absorbed in the upper jejunum
Only **monosaccharides** can be absorbed

If disaccharidase deficiency → undigested carbs in large intestine:

- **Osmotic diarrhea**: water drawn into lumen
- **Bacterial fermentation** → $\text{CO}_2 + \text{H}_2$ + short-chain acids
- Causes **abdominal cramps, bloating, and flatulence**

Type	Description
Genetic	Inherited enzyme defect (e.g., lactase, sucrase-isomaltase)
Acquired	Intestinal disease, malnutrition, mucosal injury, or severe diarrhea (temporary)



Carbohydrate Digestion

Common Disorders & Diagnosis

1. Lactose Intolerance (Lactase Deficiency):

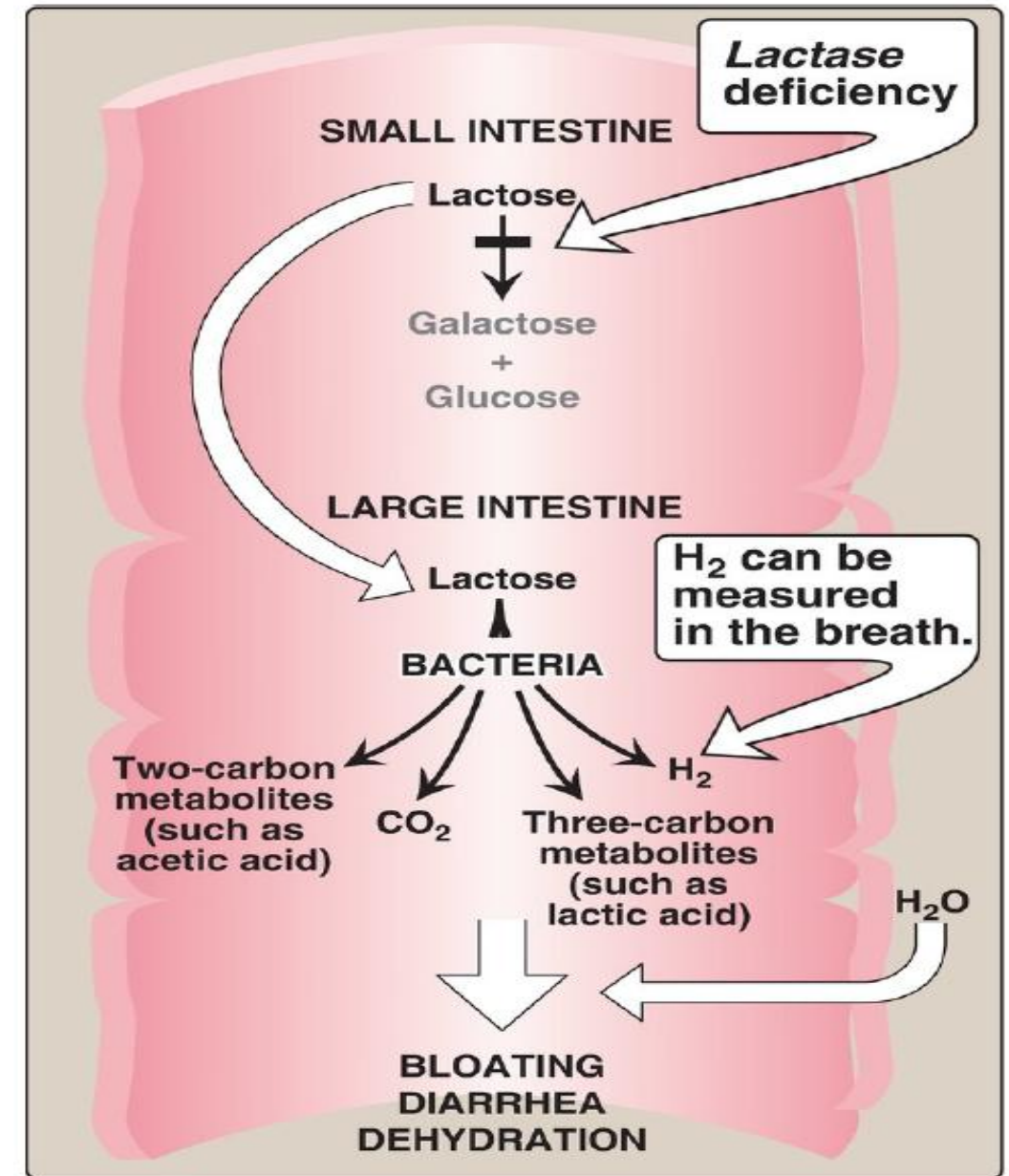
- Affects >60% of adults worldwide
- Common in **African & Asian descent** (up to 90%)
- Caused by ↓ lactase gene expression on **chromosome 2**
- **Symptoms:** Bloating, cramps, diarrhea after milk
- **Management:**
 - Limit milk, choose yogurt/aged cheese
 - Use lactase-treated products or supplements
 - Ensure **calcium** from green vegetables

2. Congenital Sucrase-Isomaltase Deficiency:

- **Autosomal recessive**, intolerance to **sucrose**
- Prevalence: 1:5,000 (Europe), 1:20 (Inuit)
- **Treatment:** Restrict sucrose, enzyme therapy

Diagnosis:

- Oral disaccharide tolerance tests
- **Hydrogen breath test** → ↑H₂ = poor carbohydrate absorption



Carbohydrate Digestion

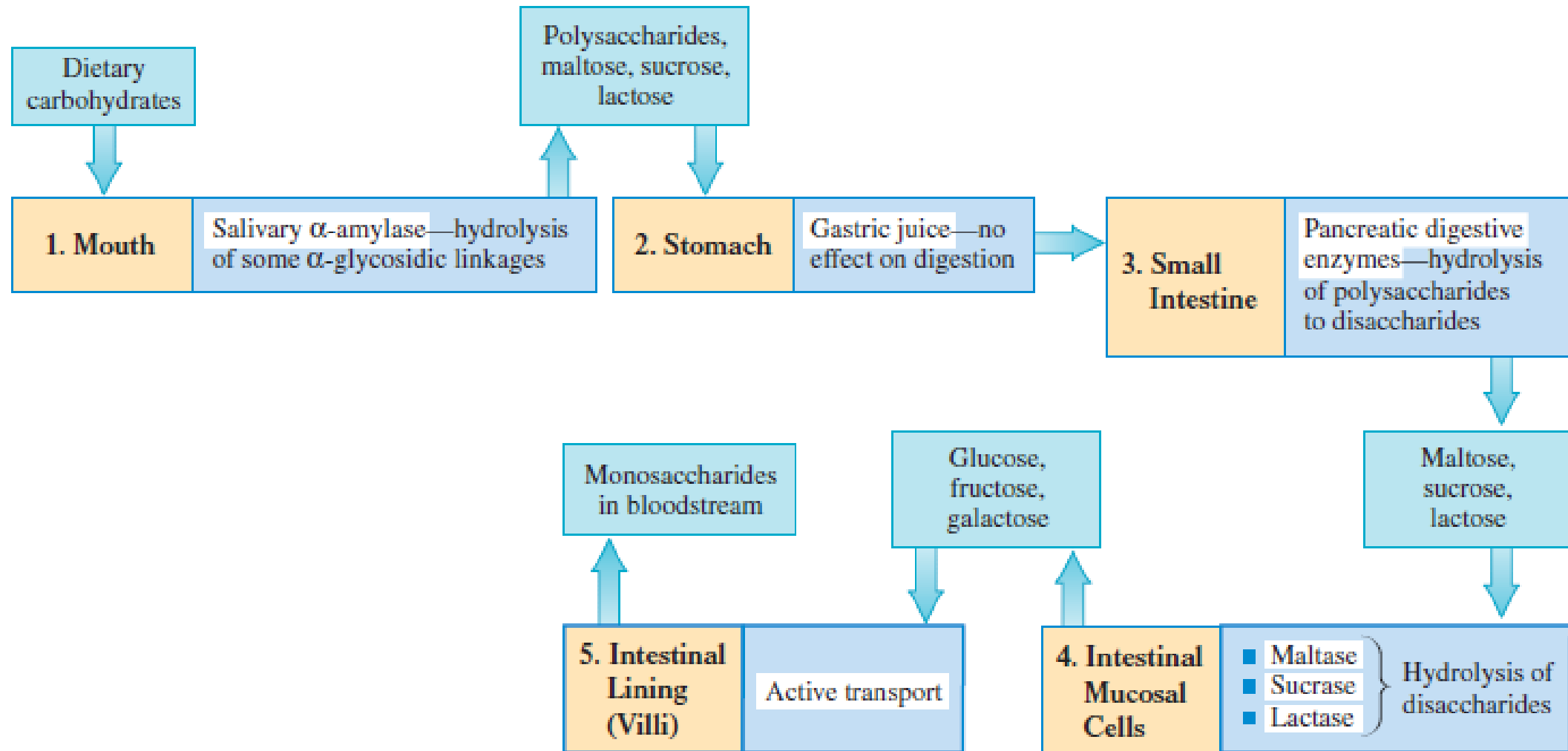
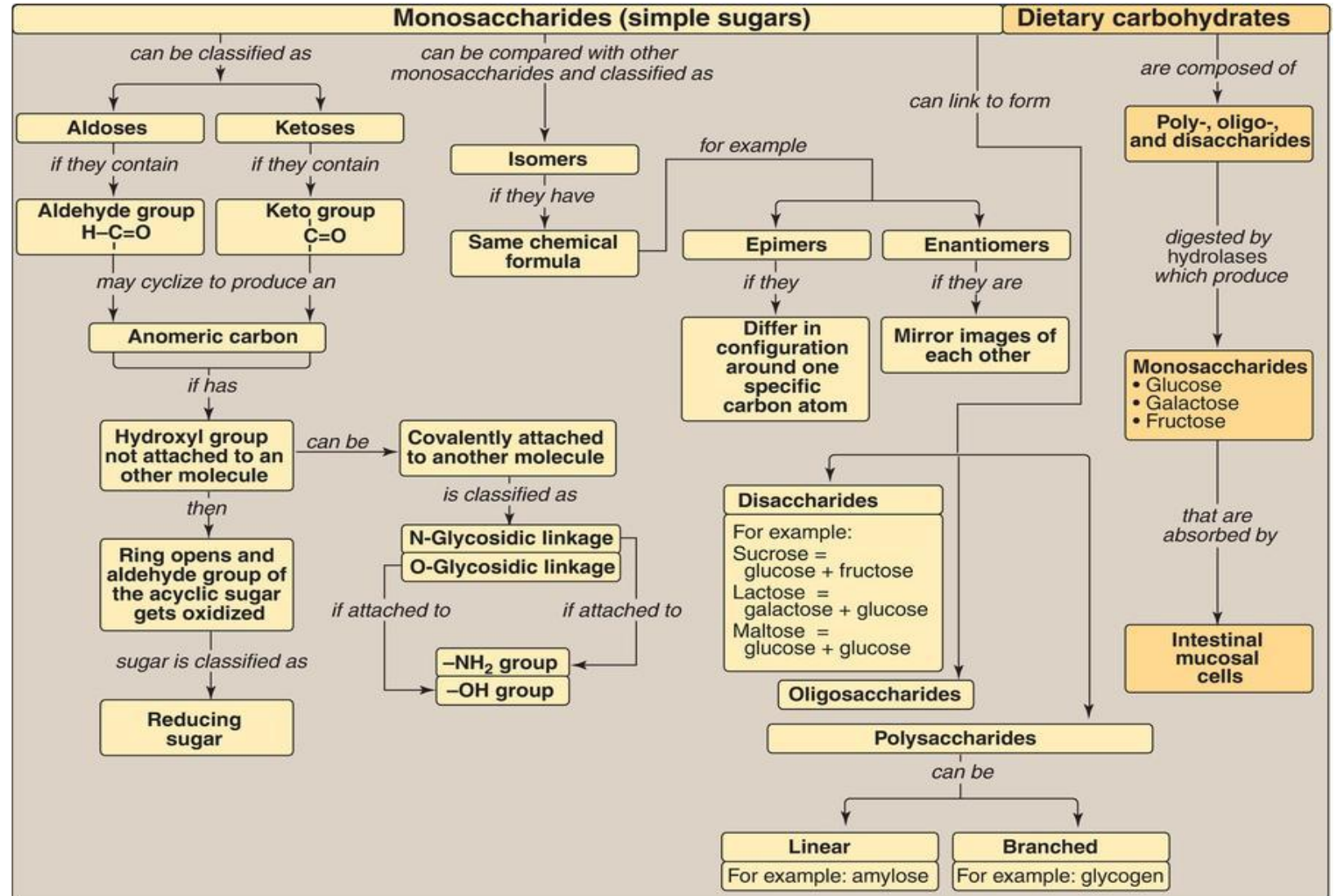


Figure 24-2 Summary of carbohydrate digestion in the human body.

Carbohydrate Digestion

Key concept map for the classification and structure of monosaccharides and the digestion of dietary carbohydrates.



Lipid Digestion

► Chyme is pronounced “kyme” (rhymes with dime).

Site	Process / Enzyme	Products Formed	Notes
Mouth	Minimal digestion (salivary enzymes ineffective)	—	TAGs are insoluble in water
Stomach	Gastric lipase hydrolyzes ~10% of TAGs	Diacylglycerols + Free fatty acids	High-fat foods → longer satiety (“feeling full”)
Small Intestine	Bile salts (from gallbladder) emulsify fat Pancreatic lipase hydrolyzes TAGs	Monoacylglycerols + Free fatty acids	Emulsification increases surface area for enzyme action

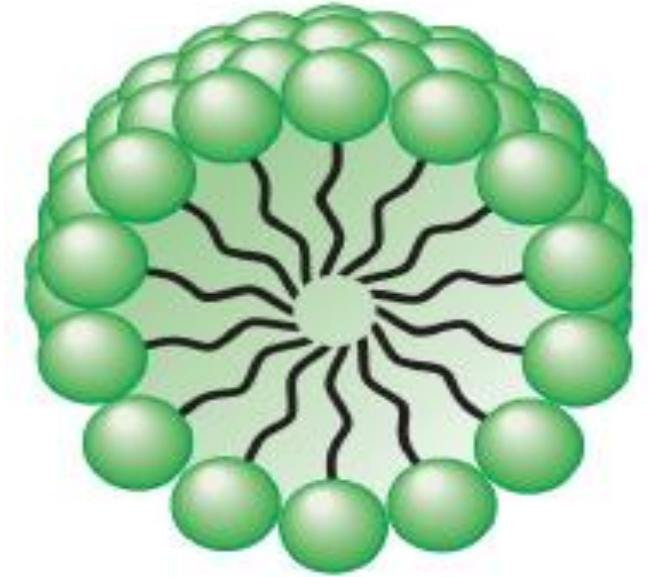


Figure 25-1 In a fatty acid micelle, the hydrophobic chains of the fatty acids and monoacylglycerols are in the interior of the micelle.

Micelle formation:

- Tiny droplets containing **fatty acids, monoacylglycerols, and bile salts**
- Make lipids **soluble for absorption** through intestinal membranes

► When freed of the triacylglycerol molecules they “transport” during digestion, bile acids are mostly recycled. Small amounts are excreted.

Lipid Digestion

► Chylomicron is pronounced
"kye-lo-MY-cron."

Inside Intestinal Cells:

- **Reassembly:** Free fatty acids + Monoacylglycerols → **TAGs**
- **Packaging:** TAGs + Cholesterol + Phospholipids + Proteins → **Chylomicrons**

Transport Pathway:

1. **Chylomicrons** enter lymphatic vessels → thoracic duct → bloodstream
2. In **blood vessels**, **lipoprotein lipase** hydrolyzes TAGs
3. **Products absorbed by tissues:**
 - **Fatty acids** → Energy (via β -oxidation)
 - **Glycerol** → Glucose synthesis or energy
 - **TAGs** → Stored in adipose tissue

Timeline:

- ↑ Chylomicrons in blood **2–6 hrs after meal**
- ↓ as TAGs stored or metabolized

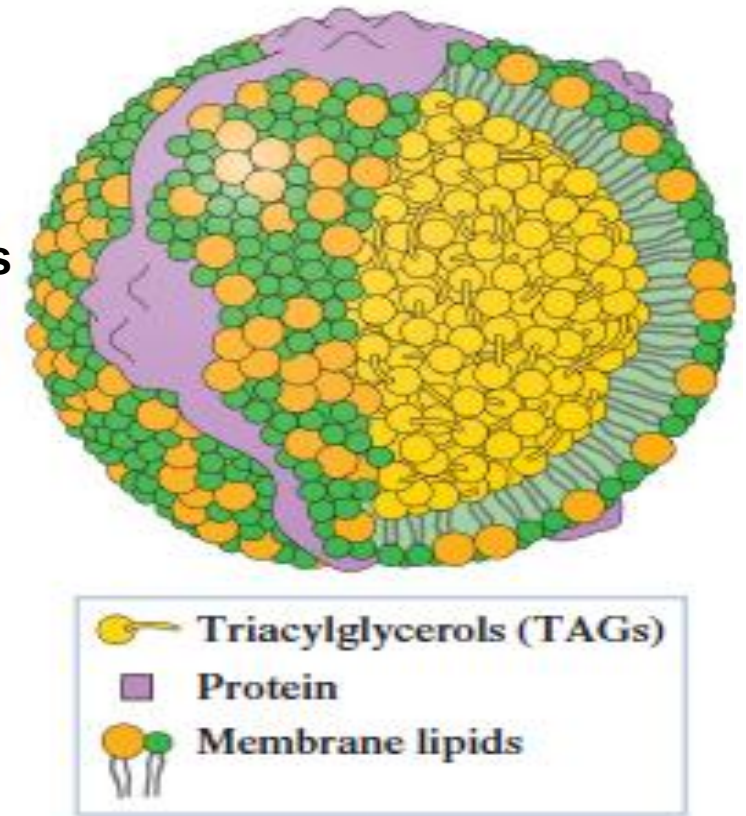


Figure 25-2 A three-dimensional model of a chylomicron, a type of lipoprotein. Chylomicrons are the form in which TAGs are delivered to the bloodstream via the lymphatic system.

Lipid Digestion

Average Lipid Intake: ~78 g/day

- >**90%** Triacylglycerols (TAGs)
- Remainder: Cholesterol, cholesteryl esters, phospholipids, free FA

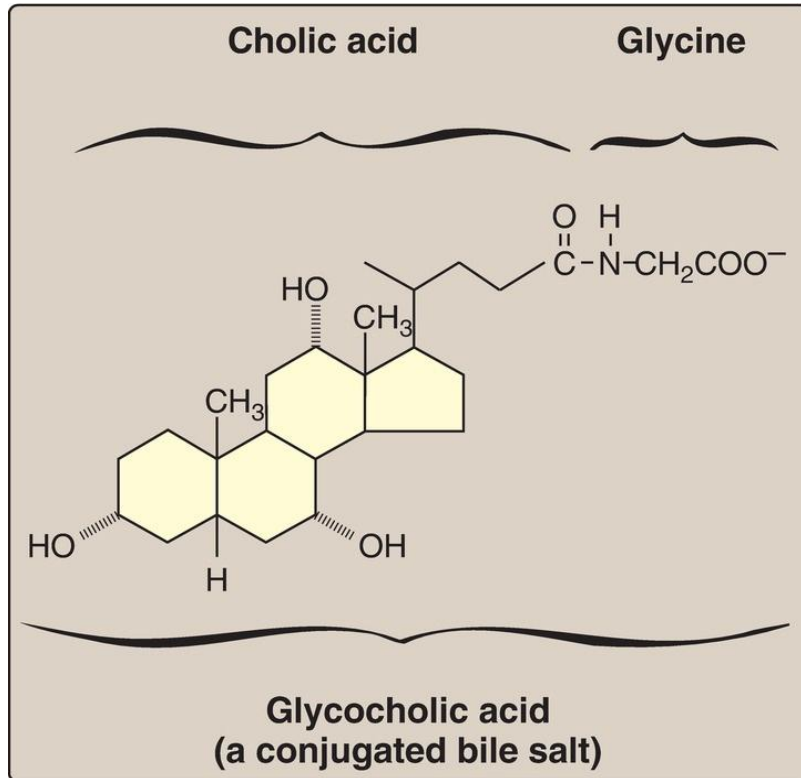
Site	Process / Enzyme	Main Products	Notes / Clinical Points
Mouth & Stomach	Lingual lipase (tongue glands) Gastric lipase (gastric mucosa)	Free FA + Diglycerides	- Active at pH 4–6 - Acts on short/medium-chain FA (milk fat) - Important for infants and CF patients
Clinical Insight	Cystic Fibrosis (CF)	↓ CFTR function → thick mucus → blocks pancreatic enzyme release	Results in pancreatic insufficiency , fat malabsorption; treated with enzyme replacement + fat-soluble vitamins



Lipid Digestion

Emulsification in the Intestine

Emulsification is the breakdown of large dietary fat droplets into smaller ones to increase surface area for enzyme action.



- Site:** Occurs in the **duodenum**.

- Purpose:** Allows **lipase enzymes** to efficiently digest lipids at the fat–water interface.

- Mechanisms:**

- 1.**Bile Salts** – Amphipathic molecules (from liver, stored in gallbladder) derived from **cholesterol** and conjugated with **glycine** or **taurine**.

- Act as natural **detergents** that coat lipid droplets.

- Prevent coalescence** and stabilize smaller droplets.

- 2.**Peristalsis** – Mechanical churning that further breaks down fat globules.

- Result:** Formation of stable, small lipid droplets ready for **lipase digestion and absorption**.

Lipid Digestion

Enzymatic Degradation & Hormonal Regulation

Degradation	Enzyme(s)	Products	Key Notes
TAGs	Pancreatic lipase + Colipase	2-Monoacylglycerol (MAG) + FFA	Colipase anchors lipase; activated by trypsin ; inhibited by Orlistat
Cholesteryl esters	Cholesterol esterase	Cholesterol + FFA	Requires bile salts for activation (INCREASED)
Phospholipids	Phospholipase A₂ → Lysophospholipase	Lysophospholipids + FFA	PA ₂ = activated by TRYPSIN Produces glycerylphosphoryl base (absorbed/excreted)



Lipid Digestion

Enzymatic Degradation & Hormonal Regulation

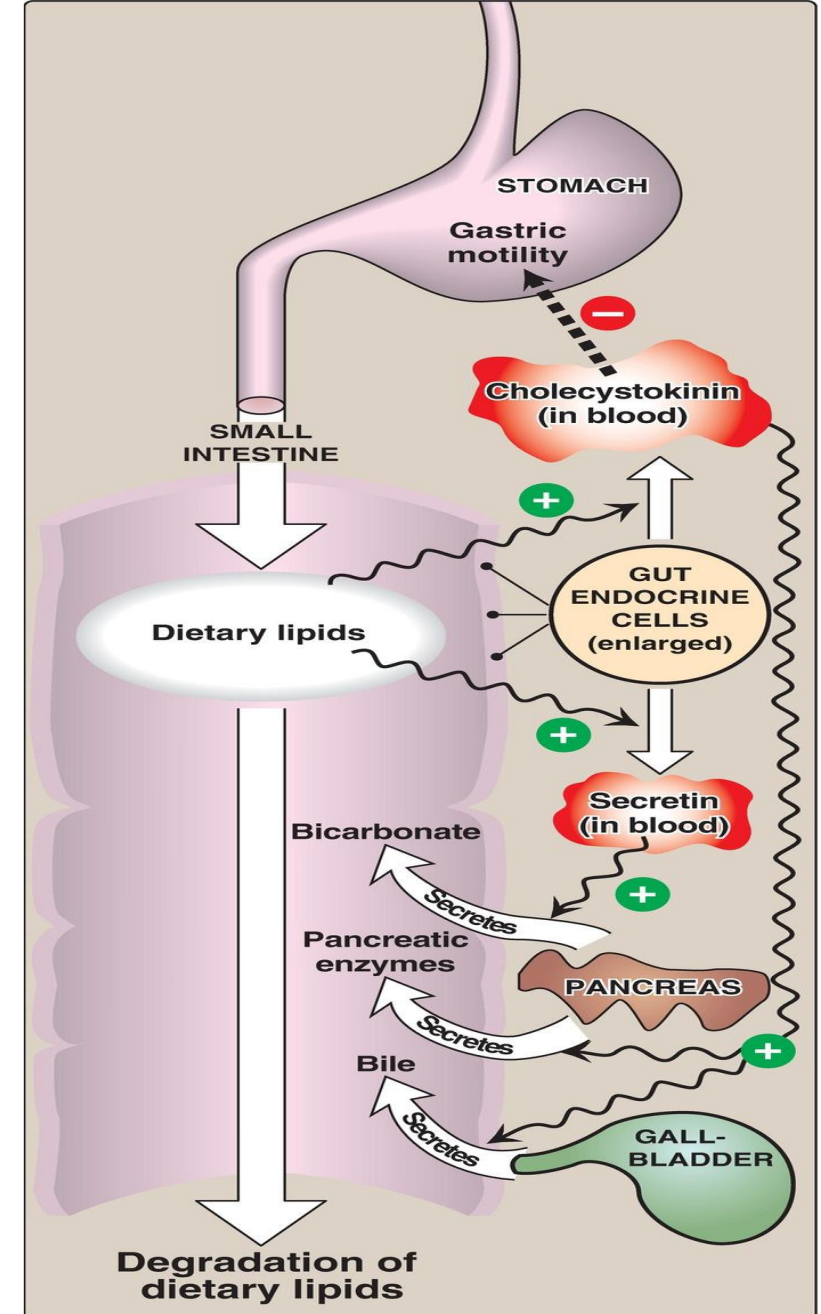
Hormonal Control of Lipid Digestion:

•Cholecystikin (CCK):

- Released in response to **lipids & peptides**
- Stimulates **gallbladder contraction** → **bile release**
- Stimulates **pancreatic enzyme secretion**
- Slows **gastric emptying**

•Secretin:

- Released in response to **acidic chyme**
- Stimulates **bicarbonate secretion** from pancreas
- Neutralizes intestinal pH for enzyme activity



Lipid Digestion

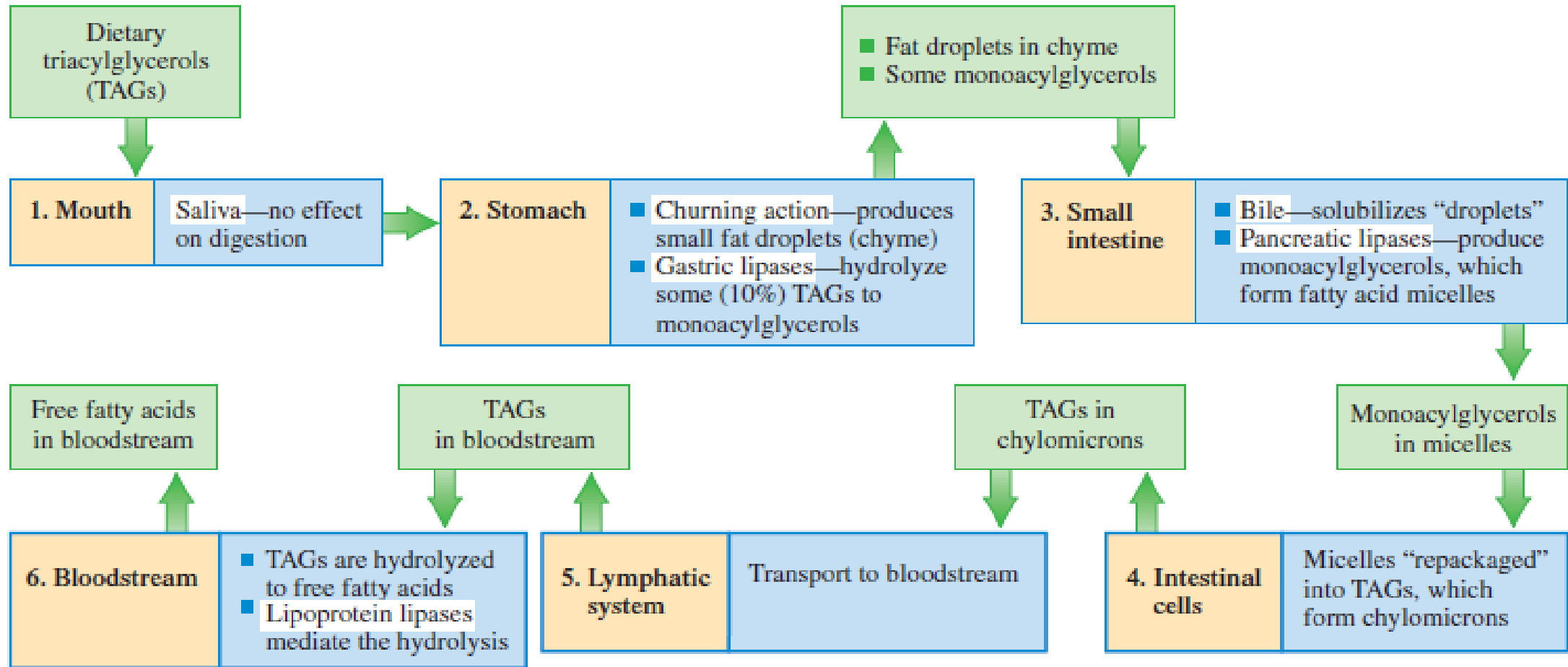
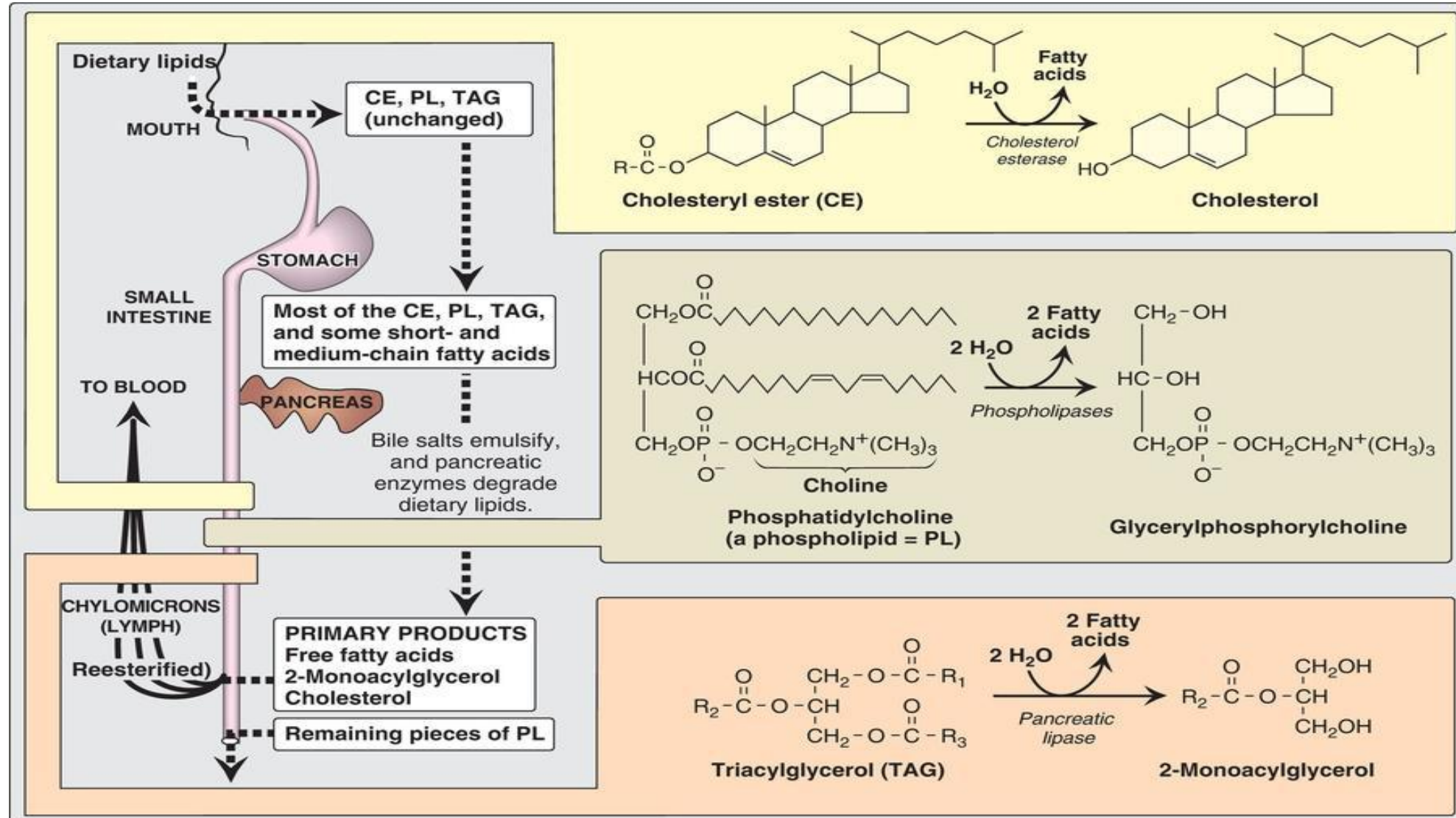
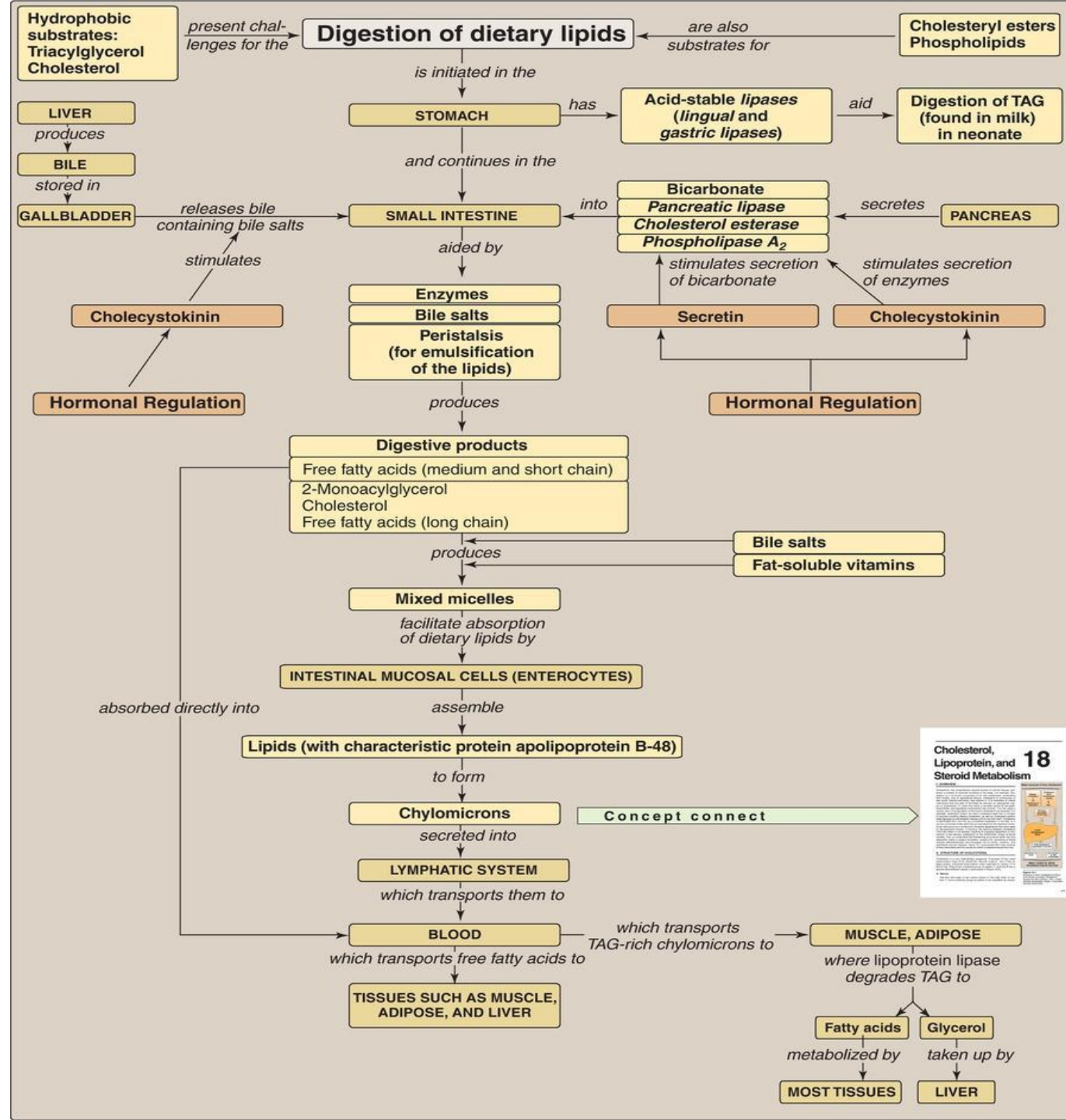


Figure 25-3 A summary of the events that must occur before triacylglycerols (TAGs) can reach the bloodstream through the digestive process.

Lipid Digestion

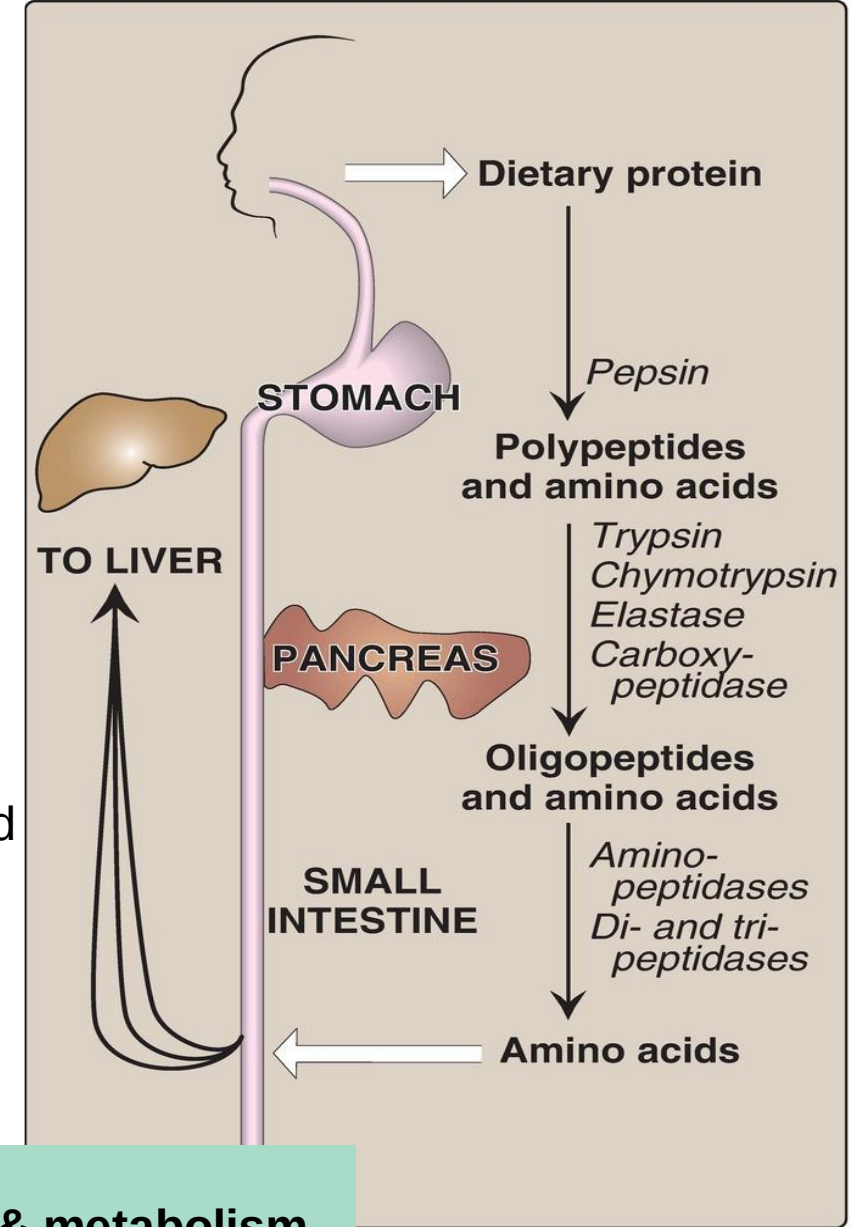




Protein Digestion

- **Starts in:** *Stomach* (no protein digestion in mouth)
- **Key Processes:**
- **In the Stomach:**
 - **Hormone Gastrin** → stimulates **HCl** and **pepsinogen** secretion
 - **HCl Functions:**
 1. Kills bacteria
 2. **Denatures proteins** (unwinds structure)
 3. **Activates pepsinogen** → **pepsin**
 - **Pepsin:** Hydrolyzes ~10% of peptide bonds → **large polypeptides**
- **In the Small Intestine:**
 - **Secretin** → triggers **pancreatic bicarbonate (HCO_3^-)** to neutralize acid
 - **pH 7–8** allows activation of **proteolytic enzymes:**
 - *Trypsin, Chymotrypsin, Carboxypeptidase, Aminopeptidase*
 - Break peptides → **amino acids**
- **Absorption:**
- **Active transport** through intestinal wall → **bloodstream**

• **Result:** Free amino acids distributed to body tissues for **protein synthesis & metabolism**



Protein Digestion

Digestion in the Stomach

- **Gastric Juice** = HCl + Pepsinogen (proenzyme)
 - **HCl (from parietal cells):**
 - Kills bacteria
 - **Denatures proteins** → exposes peptide bonds
 - *pH 2–3* (too weak to hydrolyze directly)
 - **Pepsin (from chief cells):**
 - Secreted as **inactive pepsinogen (zymogen)**
 - Activated by HCl → **pepsin**
 - **Endopeptidase** → breaks large proteins into **polypeptides & few amino acids**



Protein Digestion

Digestion by gastric secretion

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 - **HCl (from parietal cells):**
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 - Activated by HCl → **pepsin**
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Protein Digestion

Digestion by pancreatic enzyme

- **Secretin** → stimulates pancreas to secrete **bicarbonate (HCO_3^-)** → neutralizes acid
- Pancreatic **proteases** continue digestion:
 - **Endopeptidases:** trypsin, chymotrypsin, elastase
 - **Exopeptidases:** carboxypeptidases A & B
 - Each enzyme specific to certain **amino acid R-groups**

Zymogen Activation

1. **Cholecystikinin (CCK)** → triggers enzyme release
2. **Enteropeptidase (brush border)** activates **trypsinogen** → **trypsin**
3. ***Trypsin activates all other pancreatic zymogens → proteolytic cascade***

Clinical Notes

- **Pancreatic deficiency (e.g., pancreatitis, cystic fibrosis)** → incomplete digestion → **steatorrhea** (fat in stool)
- **Celiac disease:** immune reaction to gluten → villi damage → malabsorption



Protein Digestion

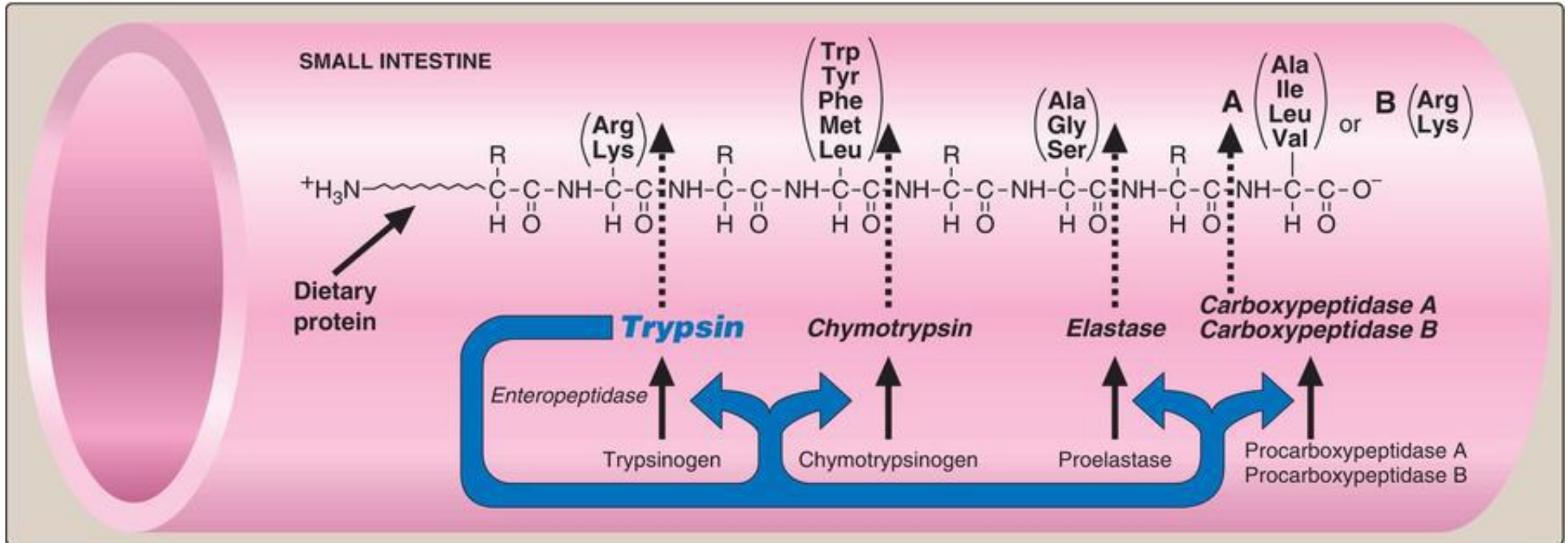


Figure 19.5 Cleavage of dietary protein in the small intestine by pancreatic *proteases*. The peptide bonds susceptible to hydrolysis are shown for each of the five major pancreatic *proteases*.

[Note: The first three are serine *endopeptidases*, whereas the last two are *exopeptidases*. Each is produced from an inactive zymogen.]

Protein Digestion

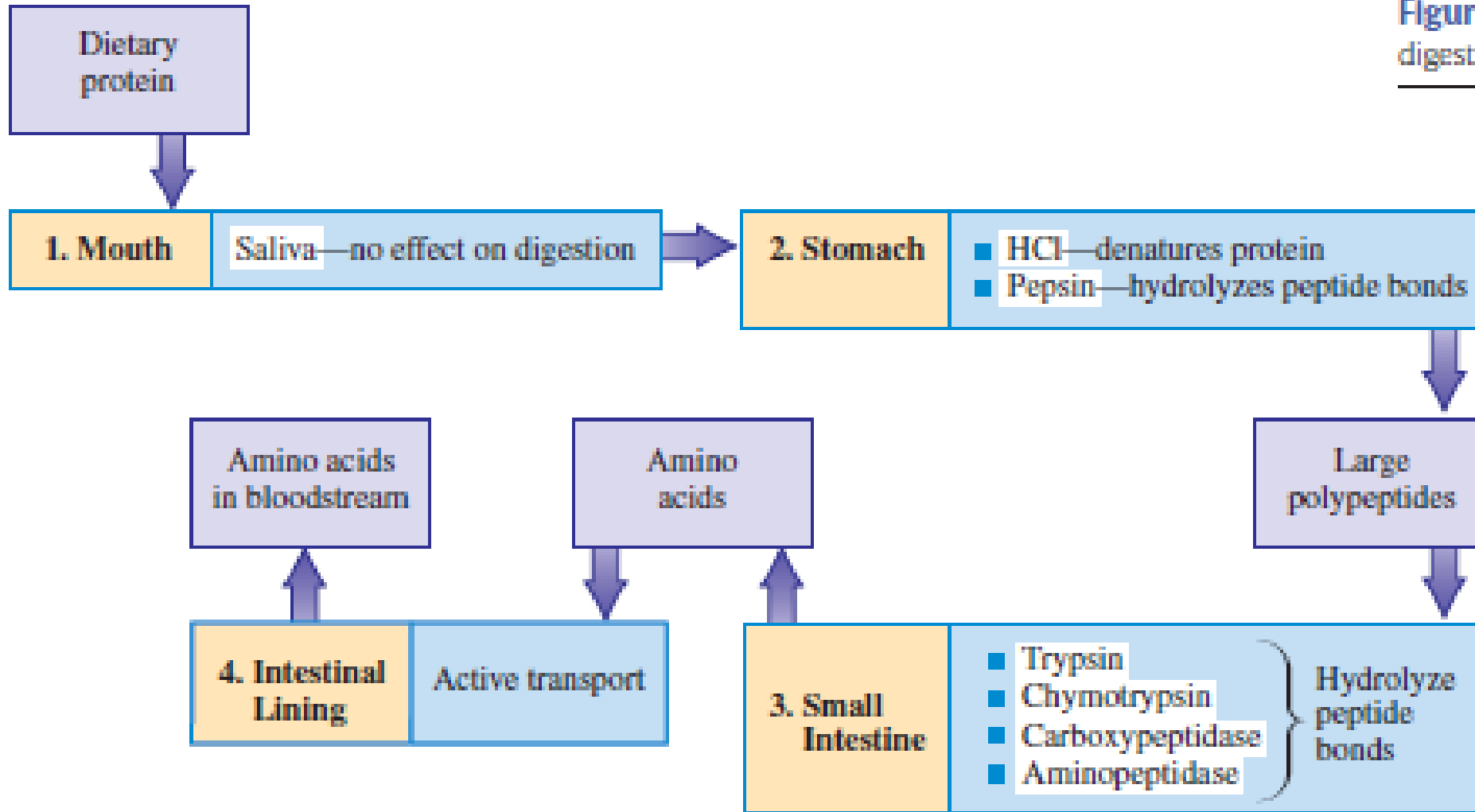


Figure 26-1 Summary of protein digestion in the human body.

Digestion

- The energy in **glucose** is used to **produce ATP** via the reactions of **glycolysis, cellular respiration, and the electron transport system.**
- **Glycerol and fatty acids** can be converted to **pyruvate** and **Acetyl CoA** and then enter **cellular respiration.**
- The body uses **amino acids** to construct proteins. Excess amino acids can be used to **synthesize pyruvate, acetyl CoA, and alpha ketogluterate**, which enters the **Krebs cycle.**



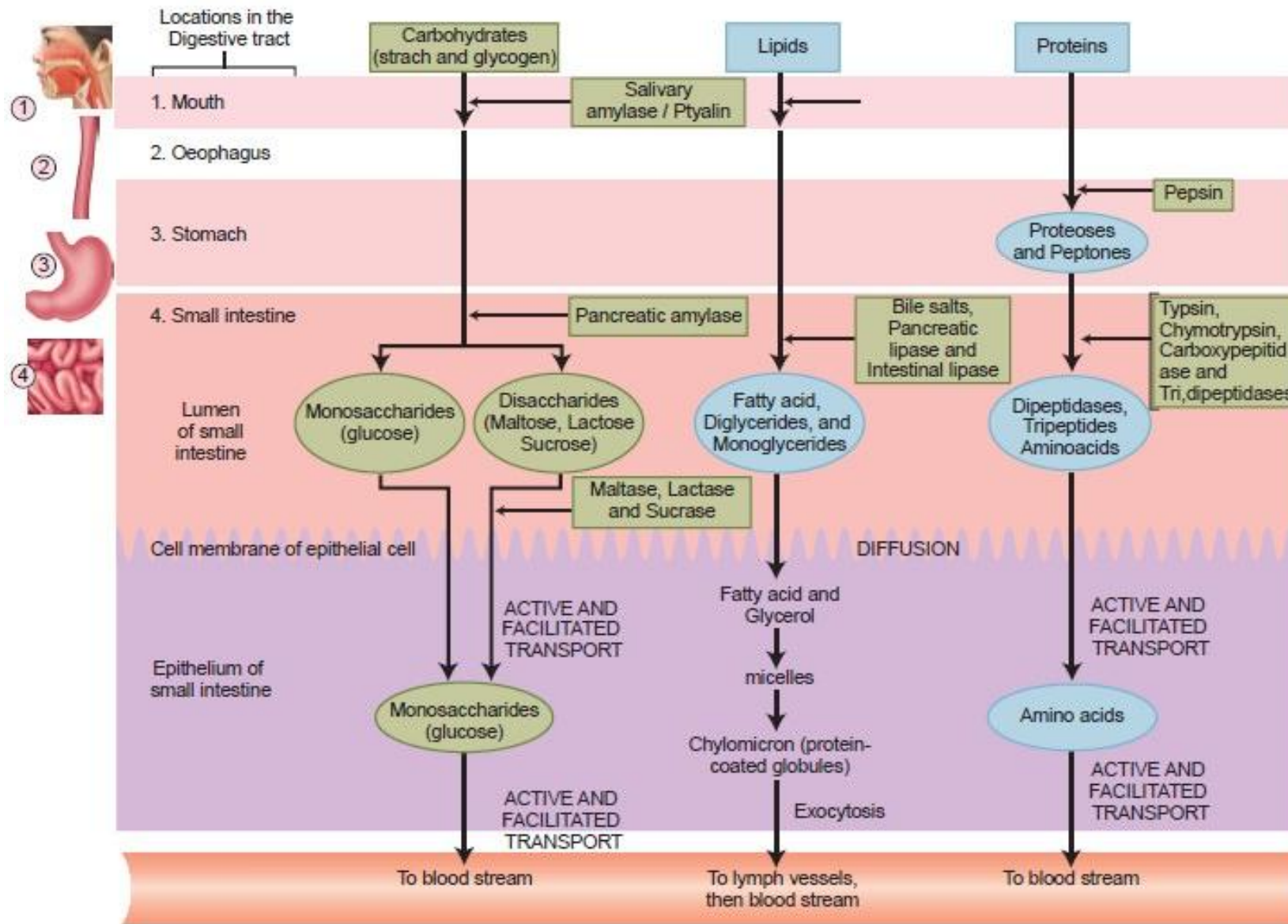
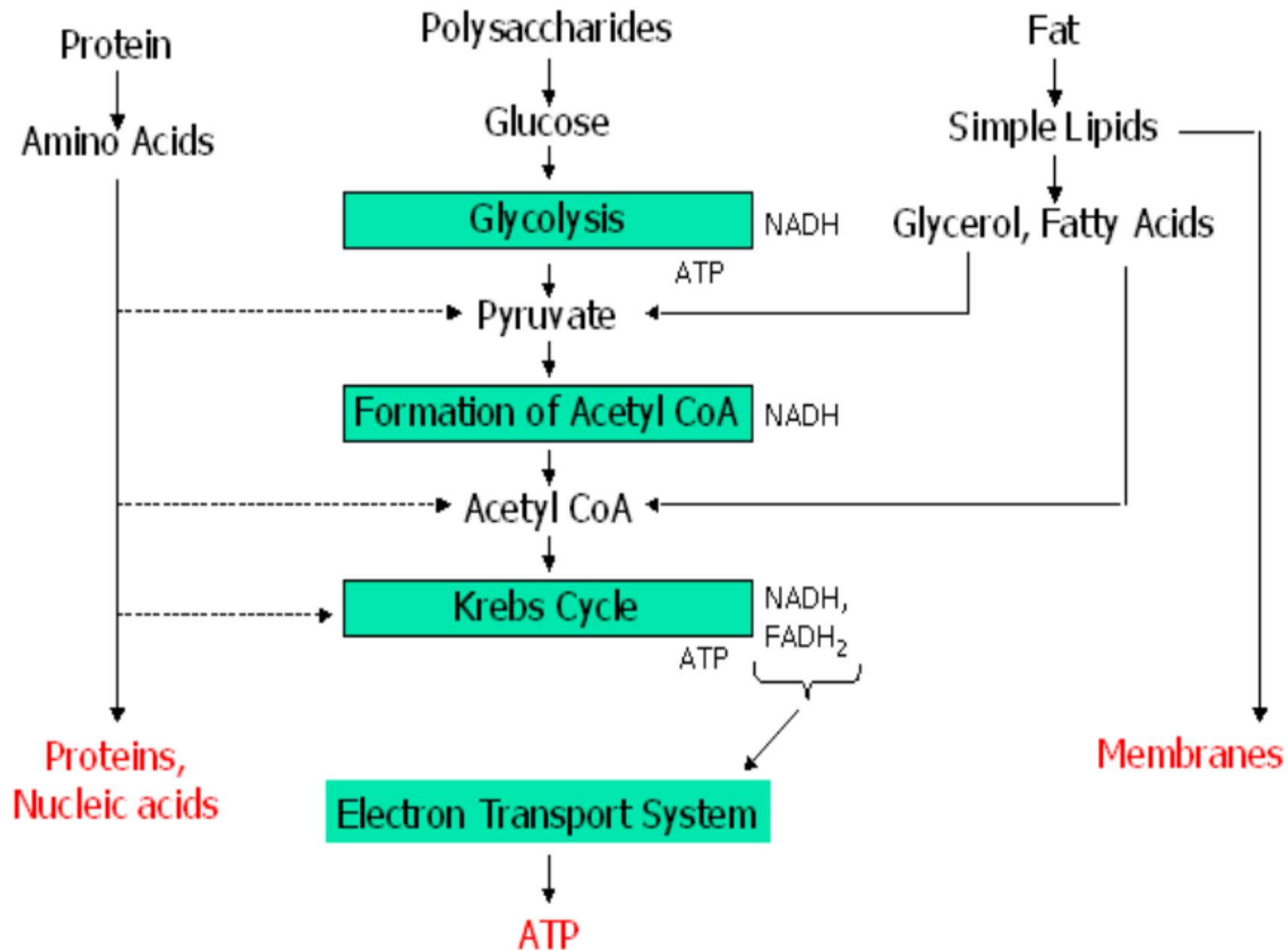
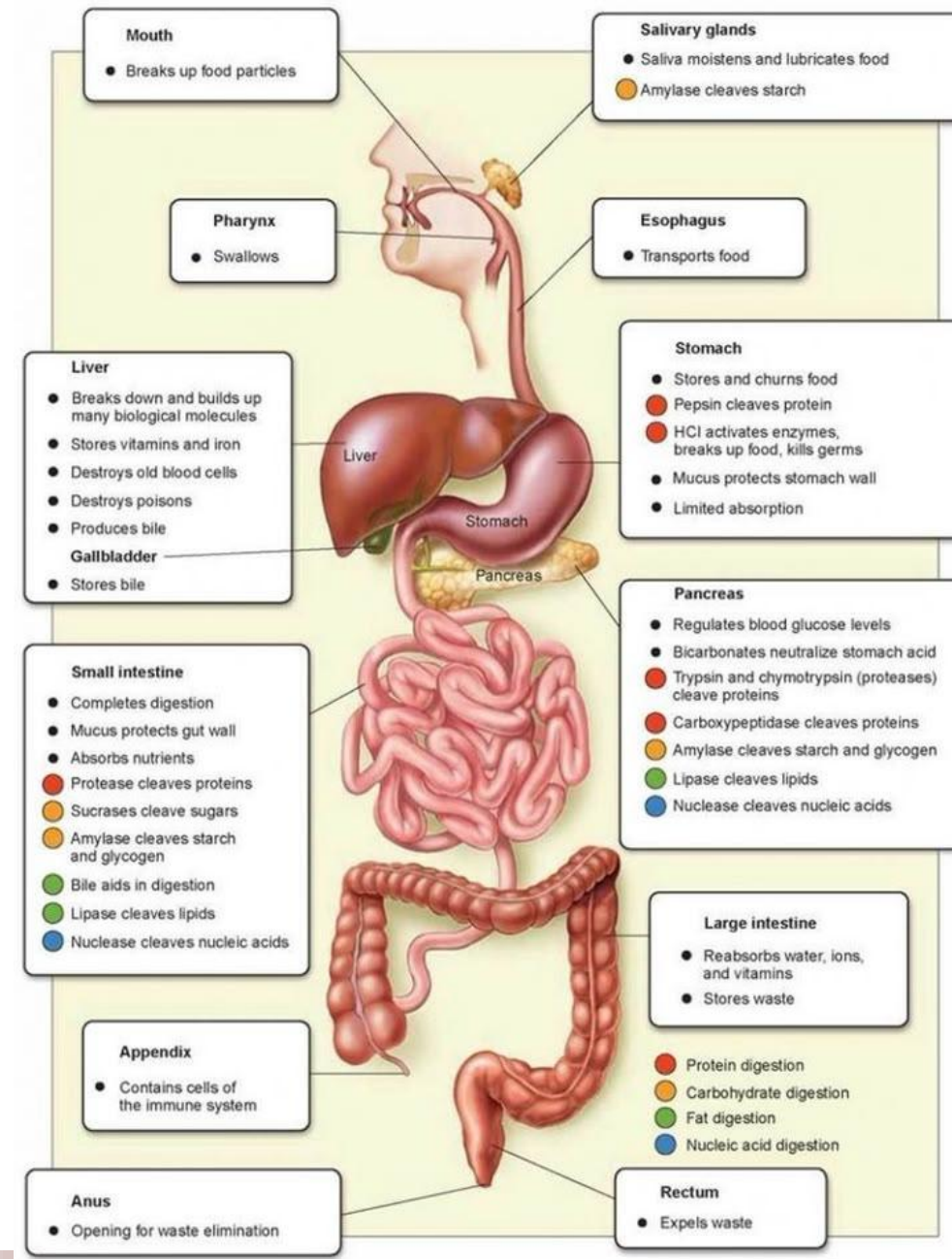


Figure 5.9 Process of Digestion and absorption



Digestion in GI Tract

- Mouth
- Stomach
- Duodenum
- Pancreas
- Liver
- The Large Intestine



Digestion in GI Tract

• Mouth

- Chewing breaks food into smaller particles so that chemical digestion can occur faster.
- Enzymes: **Salivary amylase** breaks starch (a polysaccharide) down to maltose (a disaccharide).
- Bicarbonate ions in saliva act as buffers, maintaining a pH between 6.5 and 7.5.
- Mucins (mucous) lubricate and help hold chewed food together in a clump called a bolus.



Digestion in GI Tract

- **Stomach**

- The stomach stores up to 2 liters of food. Gastric glands within the stomach produce secretions called **gastric juice**.
- The muscular walls of the stomach contract vigorously to mix food with gastric juice, producing a mixture called **chyme**.



Digestion in GI Tract

• Gastric Juice

- Pepsinogen is converted to pepsin, which digests proteins. Pepsinogen production is stimulated by the presence of gastrin in the blood.
- Hydrochloric acid (HCl) converts pepsinogen to pepsin which breaks down **proteins to peptides**.
- HCl maintains a pH in the stomach of approximately 2.0. It also dissolves food and kills microorganisms.
- Mucous protects the stomach from HCl and pepsin
- **Secretion of Gastric Juice**: Gastrin is a hormone that stimulates the stomach to secrete gastric juice



Digestion in GI Tract

- **Duodenum**

- First part of the small intestine.
- Chyme enters in tiny spurts.
- At this point, proteins and carbohydrates are only partially digested and lipid digestion has not begun.



Digestion in GI Tract

• Pancreas

- The pancreas acts as an exocrine gland by producing pancreatic juice which empties into the small intestine via a duct.
- The pancreas also acts as an endocrine gland to produce insulin.

Pancreatic juice contains sodium bicarbonate which neutralizes the acidic material from the stomach.

- **Pancreatic amylase** digests starch to maltose.
- **Trypsin** and **Chymotrypsin** digest proteins to peptides. Like pepsin (produced in the stomach), they are specific for certain amino acids, not all of them. They therefore produce peptides.
- **Lipase** digests fats to glycerol and fatty acids.



Digestion in GI Tract

- **Liver**

- The liver produces bile which is stored in gallbladder and sent to the duodenum through a duct.
- Bile emulsifies fats (separates it into small droplets) so they can mix with water and be acted upon by enzymes.



Digestion in GI Tract

• Small Intestine

- there are numerous projections called villi that function to increase the surface area of the intestine.
- Individual villus cells have microvilli = greatly increase absorptive surface area.
- Peptidases and maltase are embedded within the plasma membrane of the microvilli.
 - Peptidases complete the digestion of peptides to amino acids.
 - Maltase completes the digestion of disaccharides.



Digestion in GI Tract

- **Large Intestine**

- Functions in three processes:

- **Recovery of water and electrolytes from ingesta**

- **Formation and storage of feces:**

- As ingesta is moved through the large intestine, it is dehydrated, mixed with bacteria and mucus, and formed into feces.

- **Microbial fermentation:**

- The large intestine of all species teems with microbial life. Those microbes produce enzymes capable of digesting many of molecules that to vertebrates are indigestible, cellulose being a premier example.

- Absorption: water, sodium ions and chloride ions

- **Secretion:** bicarbonate ions and mucus

Ingesta: material taken into the body by way of the digestive tract



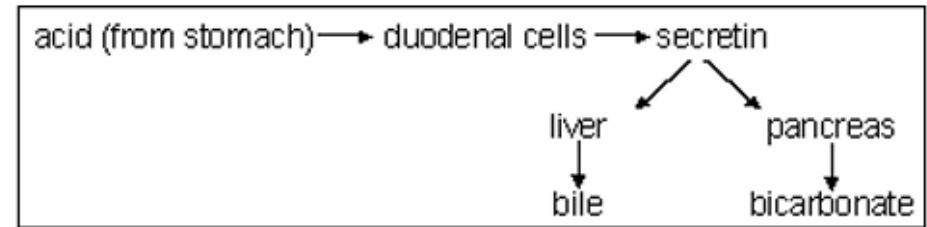
Digestion in GI Tract

• Hormones Involved Digestion

1. **Gastrin:** The presence of food in the stomach stimulates specific receptors which in turn stimulates endocrine cells in the stomach to secrete the hormone **gastrin** into the circulatory system. Gastrin stimulates the stomach to secrete gastric juice.
2. **Secretin:** Secretin is produced by cells of the duodenum.

It's production is stimulated by acid chyme from stomach.

It stimulates the



pancreas to produce sodium bicarbonate, which neutralizes the acidic chyme. It also stimulates the liver to secrete bile.

3. **CCK (cholecystokinin):** CCK production is stimulated by the presence of food in the duodenum. It stimulates the gallbladder to release bile and the pancreas to produce pancreatic enzymes.
4. **GIP (Gastric Inhibitory Peptide):** Food in the duodenum stimulates certain endocrine cells to produce GIP.

It has the opposite effects of gastrin; it inhibits gastric glands in the stomach and it inhibits the mixing and churning movement of stomach muscles. This slows the rate of stomach emptying when the duodenum contains food.



Digestion in GI Tract

- **Digestive Enzymes grouped by source**

SOURCE	ENZYME	FOOD	PRODUCT
MOUTH (salivary glands)	Salivary amylase	Polysaccharides	Maltose
STOMACH	Pepsin	Proteins	Peptides
PANCREAS	Pancreatic amylase Trypsin Lipase	Polysaccharides Proteins Fats	Maltose Peptides Fatty acids and glycerol
SMALL INTESTINE	Maltase Peptidases	Maltose Peptides	Glucose Amino acids



Overview of Digestion & Absorption

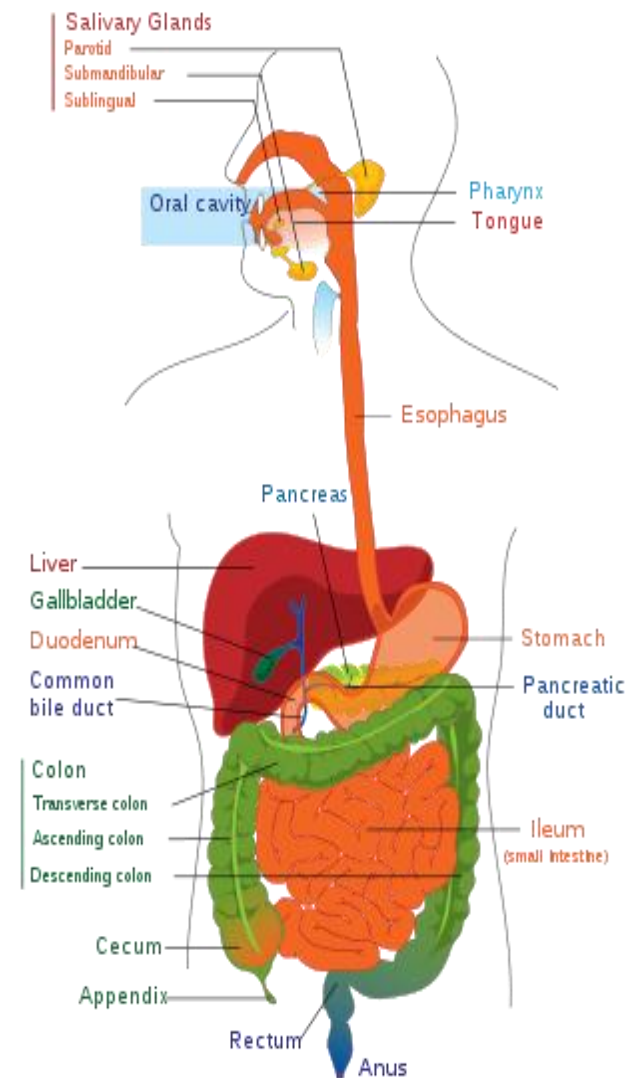


	Fiber	Carbohydrate
Mouth and salivary glands	The mechanical action of the mouth and teeth crushes and tears fiber in food and mixes it with saliva to moisten it for swallowing.	The salivary glands secrete saliva into the mouth to moisten the food. The salivary enzyme amylase begins digestion: $\text{Starch} \xrightarrow{\text{amylase}} \text{small polysaccharides, maltose.}$
Stomach	Fiber is unchanged.	Stomach acid inactivates salivary enzymes, halting starch digestion. To a small extent, stomach acid hydrolyzes maltose and sucrose
Small intestine	Fiber is unchanged.	The pancreas produces enzymes and releases them through the pancreatic duct into the small intestine: $\text{Polysaccharides} \xrightarrow[\text{amylase}]{\text{pancreatic}} \text{disaccharides.}$ Then enzymes on the surfaces of the small intestinal cells break disaccharides into monosaccharides, and the cells absorb them : $\begin{array}{lcl} \text{Maltose} & \xrightarrow{\text{maltase}} & \text{glucose} + \text{glucose.} \\ \text{Sucrose} & \xrightarrow{\text{sucrase}} & \text{fructose} + \text{glucose.} \\ \text{Lactose} & \xrightarrow{\text{lactase}} & \text{galactose} + \text{glucose.} \end{array}$
Large intestine	Most fiber passes intact through the digestive tract to the large intestine. Here, bacterial enzymes digest some fiber: $\text{Some fiber} \xrightarrow[\text{enzymes}]{\text{bacterial}} \text{fatty acids, gas.}$ Fiber holds water; regulates bowel activity; and binds cholesterol and some minerals, carrying them out of the body as it is excreted with feces	



	Fat	Protein
Mouth and salivary glands	The sublingual salivary gland in the base of the tongue secretes as salivary lipase. Some hard fats begin to melt as they reach body temperature.	Chewing and crushing moisten protein-rich foods and mix them with saliva to be swallowed.
Stomach	The acid-stable salivary lipase splits one bond of triglycerides to produce diglycerides and fatty acids. The stomach's churning action mixes fat with water and acid. A gastric lipase accesses and hydrolyzes a very small amount of fat.	Hydrochloric acid (HCL) uncoils protein strands and activates stomach enzymes: $\text{Protein} \xrightarrow[\text{HCl}]{\text{pepsin}} \text{smaller polypeptides.}$
Small intestine	Bile flows in from the liver and gallbladder (via the common bile duct): $\text{Fat} \xrightarrow{\text{bile}} \text{emulsified fat.}$ Pancreatic lipase flows in from the pancreas (via the pancreatic duct): $\text{Emulsified fat} \xrightarrow[\text{lipase}]{\text{pancreatic}} \text{monoglycerides, glycerol, fatty acids (absorbed).}$	$\text{Polypeptides} \xrightarrow[\text{and intestinal proteases}]{\text{pancreatic}} \text{tripeptides, dipeptides, amino acids}$ Then enzymes on the surface of the small intestinal cells hydrolyze these peptides and the cells absorb them. $\text{Peptides} \xrightarrow[\text{dipeptidases and tripeptidases}]{\text{intestinal}} \text{amino acids (absorbed)}$
Large intestine	Some fat and cholesterol, trapped in fiber, exit in feces.	

	Vitamin	Water and Minerals
Mouth and salivary glands	No action.	The salivary glands add water to disperse and carry food.
Stomach	Intrinsic factor attaches to vitamin B12.	Stomach acid (HCl) acts on iron to reduce it, making it more absorbable. The stomach secretes enough watery fluid to turn a moist, chewed mass of solid food into liquid chyme.
Small intestine	Bile emulsifies fat-soluble vitamins and aids in their absorption with other fats. Water-soluble vitamins are absorbed.	The small intestine, pancreas, and liver add enough fluid so that approximately 2 gallons are secreted into the intestine in a day. Many minerals are absorbed. Vitamin D aids in the absorption of calcium.
Large intestine	Bacteria produce vitamin K, which is absorbed.	More minerals and most of the water are absorbed.



Digestion Process

- To digest food, five different body organs secrete digestive juices: the salivary glands, the stomach, the small intestine, the liver (via the gallbladder), and the pancreas.
- These secretions enter the GI tract at various points along the way, bringing an abundance of water and a variety of enzymes. Each of the juices has a turn to mix with the food and promote its breakdown to small units that can be absorbed into the body.

DIGESTION IN THE MOUTH

- Digestion of carbohydrate begins in the mouth, where the **salivary glands** secrete **saliva**, which contains water, salts, and enzymes (including salivary **amylase**) that break the bonds in the chains of starch.
- The enzymes in the mouth do not affect the fats, proteins, vitamins, minerals, and fiber that are present in the foods people eat.



Digestion Process

DIGESTION IN THE STOMACH

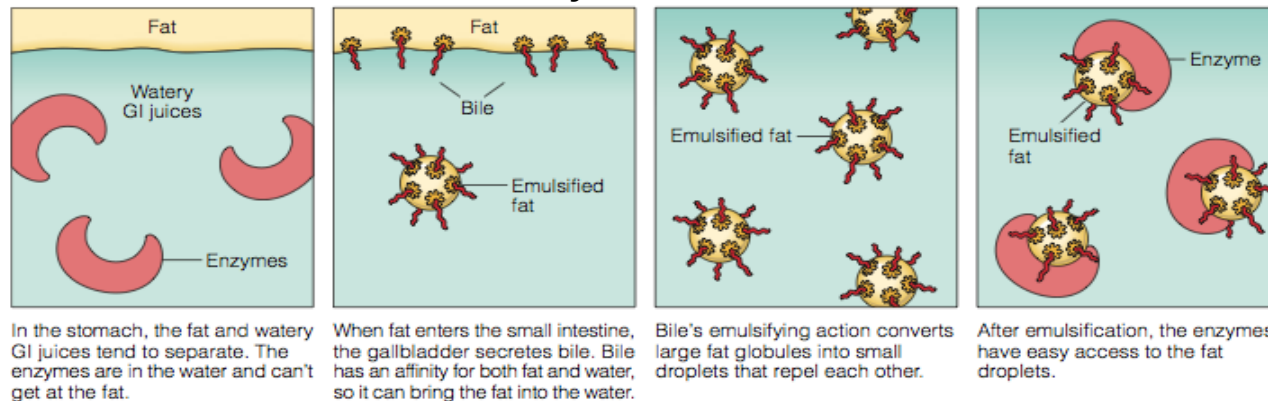
- **Gastric juice**, secreted by the **gastric glands**, is composed of water, enzymes, and **hydrochloric acid**.
- The strong acidity of the stomach prevents bacterial growth and kills most bacteria that enter the body with food.
- The strong acidity wouldn't harm the stomach itself, since the cells of the stomach wall secrete **mucus**, a thick, slimy, white polysaccharide that coats and protects the stomach's lining.
- The major digestive event in the stomach is the initial breakdown of proteins. There, the acid helps to denature the protein so that the stomach enzymes can attack the bonds.
- Both the enzyme **pepsin** and the stomach acid itself act as catalysts in the process. Minor events are the digestion of some fat by a gastric lipase, the digestion of sucrose (to a very small extent) by the stomach acid, and the attachment of a protein carrier to vitamin B12.



Digestion Process

DIGESTION IN THE SMALL INTESTINE

- The pancreas and the liver contribute additional digestive juices through the duct leading into the duodenum, and the small intestine adds **intestinal juice**. These juices contain digestive enzymes, bicarbonate, and bile.
- The pancreatic juice also contains sodium **bicarbonate**, which neutralizes the acidic chyme as it enters the small intestine. From this point on, the contents of the digestive tract are neutral or slightly alkaline. The enzymes of both the intestine and the pancreas work best in this environment.
- **Bile** is secreted by the **liver** and is concentrated and stored in the gallbladder.
- Bile is not an enzyme but an **emulsifier** that brings fats into suspension in water. After the fats are emulsified, enzymes can work on them, and they can be.



Like bile, detergents are emulsifiers and work the same way, which is why they are effective at removing grease spots from clothes. Molecule by molecule, the grease is dissolved out of the spot and suspended in water, where it can be rinsed away.

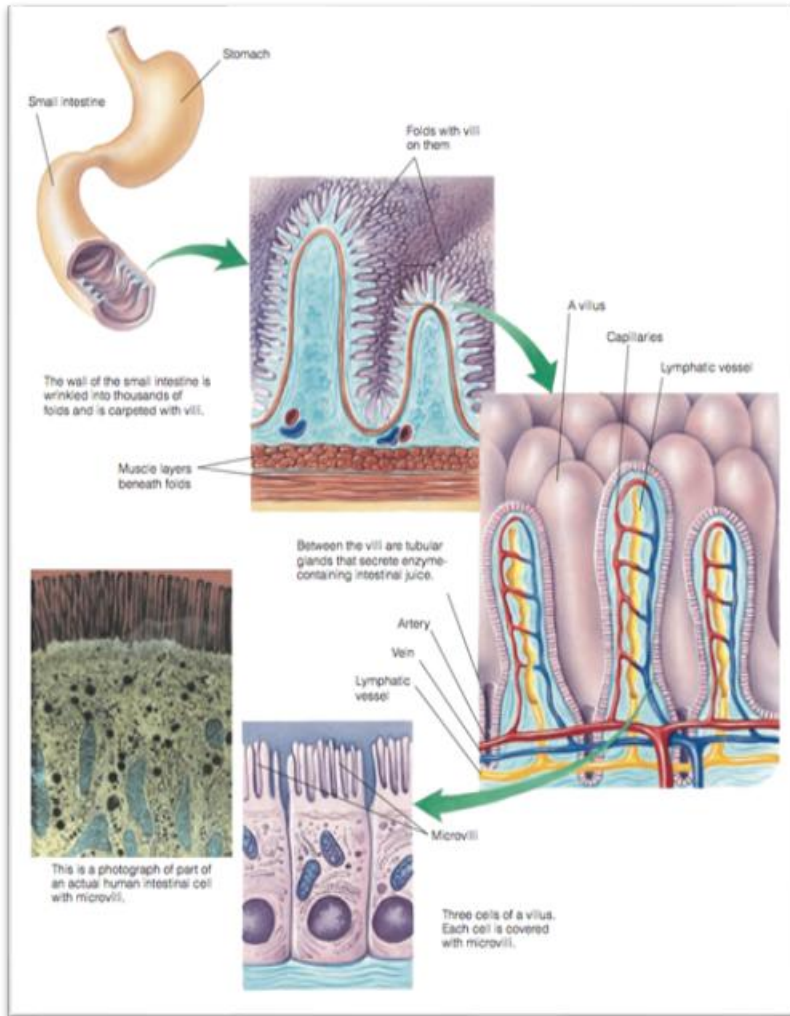
Digestion Process

DIGESTION IN THE LARGE INTESTINE

- Undigested residues, such as some fibers, are not absorbed but continue through the digestive tract as a semisolid mass that stimulates the tract's muscles, helping them remain strong and able to perform peristalsis efficiently.
- Fiber also retains water, keeping the stools soft, and carries some bile acids, sterols, and fat out of the body.



Absorptive System



- Most absorption takes place in the small intestine.
- Small intestine's inner surface looks smooth, but viewed through a microscope, it turns out to be wrinkled into hundreds of folds.
- Each fold is covered with thousands of fingerlike projections called **villi**.
- A single villus, magnified still more, turns out to be composed of several hundred cells, each covered with microscopic hairs called **microvilli**.

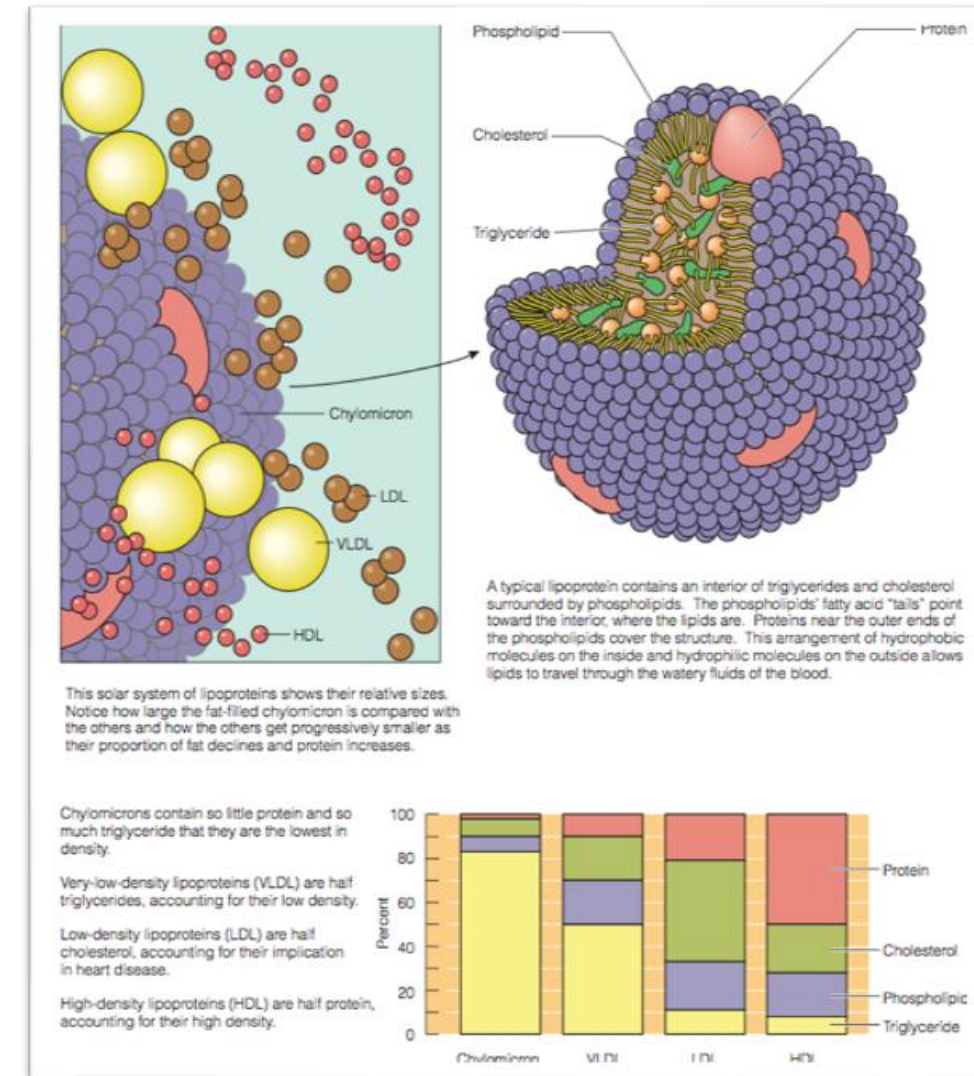
Absorptive System

- Once a molecule has entered a cell in a villus, the next step is to transmit it the **bloodstream** and the **lymphatic system**.
- Both systems supply vessels to each villus. Through these vessels, the nutrients leave the cell and enter either the **lymph** or the blood. In either case, the nutrients end up in the blood.
- The water- soluble nutrients (and the smaller products of fat digestion) are released directly into the bloodstream by way of the capillaries, but the larger fats and the fat-soluble vitamins find direct access into the capillaries impossible because these nutrients are insoluble in water (and blood is mostly water). They require some packaging before they are released.
- The intestinal cells assemble the products of fat digestion into larger molecules called **triglycerides**, which then packaged for transport.
- They cluster together with special proteins to form **chylomicrons**, one kind of **lipoproteins**. Finally, the cells release the chylomicrons into the lymphatic system.



Transport of Lipids: LIPOPROTEINS

- Within the circulatory system, lipids always travel from place to place bundled with protein, that is, as lipoproteins.
- VLDL, LDL, HDL, and chylomicrons transport newly absorbed lipids from the intestinal cells to the rest of the body.
- The liver can assemble different lipoproteins, which are known as **very-low-density lipoproteins (VLDL)**. As the body's cells remove triglycerides from the VLDL, the proportions of their lipid and protein contents shift. As this occurs, VLDL become cholesterol-rich **low-density lipoproteins (LDL)**.
- Cholesterol returning to the liver for metabolism or excretion from other parts of the body is packaged in lipoproteins known as **high-density lipoproteins (HDL)**.



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THANK YOU

Ryan Christopher G. Atuel, RPh, MSc

ryanchristopher.atuel@cit.edu.ph

Department of Pharmacy

College of Nursing and Allied Health Sciences

Cebu Institute of Technology University

Cebu, Philippines

